

Robust tube-based economic model predictive control of nonlinear systems using linear parameter varying approach

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ABSTRACT

Model Predictive Control (MPC) has become a powerful framework for optimizing system performance under constraints. However, when applied to nonlinear systems subject to unknown but bounded uncertainties (UBB), conventional MPC approaches face significant challenges related to computational complexity and robustness. This paper proposes a robust tube-based economic model predictive control (REMPC) method using a linear parameter-varying (LPV) approach for nonlinear systems with unknown but bounded uncertainty, based on nominal predictions. Using a nonlinear embedding approach, the nonlinear model is transformed into an LPV model. The optimal states and inputs found from solving the previous optimization problem are used to estimate the scheduling variables along the prediction horizon while executing the receding horizon strategy. This approach converts the nonlinear optimization problem into a quadratic optimization problem, effectively reducing computational time by leveraging the efficiency inherent in the LPV formulation. Recursive feasibility and input-to-state stability are guaranteed. Recursive feasibility is ensured by tighter constraints, which are computed online using a zonotopic approach based on the disturbance reachable sets. A gain-scheduling H_∞ controller is employed as the local controller to further tighten these constraints. The stability of the proposed approach is ensured by forcing the terminal state to converge towards the optimal equilibrium or working point of the system. Moreover, the terminal constraint is relaxed by using a constraint set around the terminal state instead of a constraint value and adding a penalty on the terminal state in the cost function. Additionally, strict dissipativity is established as a sufficient condition to prove stability. Finally, the effectiveness of the LPV-based REMPC strategy is demonstrated by controlling an isothermal Continuous Stirred Tank Reactor (CSTR), and an REMPC-LPV-based planning approach for a 1/10 scale autonomous remote-controlled (RC) electric car is also tested through simulations.

1. Introduction

Model Predictive Control has become a dominant approach in the field of advanced control strategies due to its ability to handle multivariable control problems and constraints effectively. By solving an optimization problem at each iteration, MPC can predict future system behavior and compute control actions that optimize performance. This predictive capability makes MPC highly suitable for a wide range of applications, including process control, automotive systems, and energy management [1].

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Despite its advantages, traditional MPC can be sensitive to model uncertainties and external disturbances, which may degrade performance or even destabilize the system [2]. To address these issues, robust MPC (RMPC) approaches have been developed [3]. One prominent method within this realm is robust tube-based MPC (RTMPC) [4]. This approach enhances the robustness of MPC by constructing a “tube” around a nominal trajectory, within which all possible system trajectories are guaranteed to remain despite uncertainties. This ensures that the control system remains stable and performs well even in the presence of disturbances and modeling errors.

Parallel to the development of RMPC techniques, the field of economic MPC (EMPC) has gained attention. Unlike conventional MPC, which typically focuses on tracking a reference trajectory or maintaining system stability, EMPC directly incorporates economic performance criteria into the control objective. This allows optimization of operational costs, energy usage, or other economic factors, making it particularly valuable in industries where efficiency and cost-effectiveness are paramount [5,6].

Combining the principles of robust TMPC with the objectives of EMPC leads to robust tube-based economic model predictive control (RTEMPC). This advanced approach aims to take advantage of the robustness against uncertainties provided by the tube-based method while simultaneously optimizing economic performance. Recent research in this area has shown promising results, demonstrating that RTEMPC can effectively manage trade-offs between robustness and economic efficiency [7,8].

To better understand the development and current state of the RTEMPC, it is essential to review the existing literature.

In [7], a novel robust periodic economic MPC using interval arithmetic is proposed. This approach ensures robust constraint satisfaction by considering unknown-but-bounded disturbances, leading to improved operational management and control efficiency in dynamic systems.

In [8], a RTEMPC scheme is introduced for a Linear Time-Invariant (LTI) that uses a periodicity constraint to ensure that the controlled system operates in a periodic manner, optimizing economic performance over a set of periodic trajectories without the need for a known periodic steady-state trajectory. This approach incorporates robust constraint satisfaction techniques, recursive feasibility guarantees, and an optimality certificate to verify the convergence of the closed-loop trajectory to an optimal nominal periodic steady trajectory.

In [9], a RTEMPC approach is introduced for an LTI system. The control strategy is designed to address challenges related to robust stability, convergence to the optimal operating point, and practical stability guarantees in the presence of disturbances.

[10] presents a homothetic RTEMPC approach by integrating a moving horizon estimator. The method ensures the recursive feasibility and asymptotic stability of the closed-loop system in the presence of disturbances and noise.

A limitation of the latter approaches is that they are designed for LTI systems which may pose drawbacks in accurately capturing complex nonlinear system dynamics.

Reference [11] introduces a method to deal with disturbed nonlinear systems subject to communication delays. The drawbacks of the proposed robust EMPC strategy include the use of a Min-Max approach, which can lead to conservative solutions and computational complexity. Additionally, direct prediction using nonlinear models may increase computational burden impacting control performance.

The primary contribution of this paper builds upon and extends the findings of a prior study on economic model predictive control for nonlinear systems based on the LPV approach [5]. The novelty of this paper is the introduction of a robust tube-based economic model predictive control strategy for nonlinear systems, utilizing a linear parameter varying approach (RTEMPC-LPV). The nonlinear system in question is subject to additive disturbances that are unknown but bounded. This novel robust economic MPC formulation employs the nominal system as the prediction model. This algorithm employs an LPV model to forecast future behavior. The core concept of the LPV modeling approach is that it represents the nonlinear prediction model as a mixture of linear models that rely on certain scheduling variables, thus eliminating the need for linearization [12]. These LPV-scheduling variables are derived by using the shifted outcome from the previous optimal iteration. Subsequently, the model is discretized, and a Quadratic Programming (QP) problem is solved with the aid of a numerical solver [13]. The computational time is significantly reduced compared to its nonlinear counterpart [14]. To ensure robust performance and stability in the design of tube-based EMPC for nonlinear systems, This study adopts an LPV approach. Specifically, conditions are established to guarantee recursive feasibility and input-to-state stability [15–18]. Furthermore, the research focuses on employing a robust bounding method to effectively manage uncertainty propagation within the LPV framework. Zonotopes are employed to encapsulate the error dynamics through reachable sets, ensuring comprehensive coverage of potential deviations, while offering significant computational advantages due to their closure under linear transformations, Minkowski sums, and Pontryagin differences. Unlike polyhedral or vertex-based representations, zonotopic operations scale linearly with the number of generators, allowing reachability and invariant set computations whose computational cost can be reduced by an order of magnitude compared to conventional methods [19]. The utilization of a class $\mathcal{K}$ function allows us to systematically quantify uncertainty bounds, facilitating the robust handling of disturbances and parameter estimation errors. This theoretical framework not only enhances the reliability and robustness of control strategies but also provides a solid foundation for practical implementation in diverse engineering applications requiring stringent performance guarantees. A robust control gain, specifically an LPV- H_∞ gain, is employed to tighten constraints throughout the prediction horizon by accounting for the potential impact of disturbances on the closed-loop system. This gain is designed to mitigate the influence of uncertainties, ensuring robust performance and stability. Importantly, this control gain is not required to match the terminal controller's gain. Instead, this terminal controller offers flexibility and can be independently tuned to expand the domain of attraction. This flexibility allows for optimizing the control strategy to achieve a broader operational range, thereby enhancing the overall robustness and effectiveness of the control system. Furthermore, we present a novel method for computing the terminal cost and defining the terminal region. These conditions are derived by transforming the quadratic stability requirements and performance bounds into a set of Linear Matrix Inequalities

Table 1
Comparison with related works.

Criteria	Proposed	[4]	[2]	[5]	[9]	[8]	Remark
System Type	LPV	LTI	Nonlinear	LPV	LTI	LTI	LPV handles parameter variations.
Uncertain Model	Yes	Yes	No	No	Yes	Yes	Uncertainty enhances robustness.
Economic Objective	Yes	No	No	Yes	Yes	Yes	Economic objectives improve efficiency.
Feedback Structure	Tube-based LPV- H_∞	Local law	MPC feedback	MPC feedback	Tube-based Classical RPI sets	Local law Zonotopic RPI sets	H_∞ provides robustness.
Uncertainty Propagation	Zonotopic RPI sets	Classical RPI sets	Not Applicable	Not Applicable			Zonotopes offer higher flexibility.
Computational Time	Fast	Moderate	Slow	Fast	Moderate	Moderate	Faster methods preferred for real-time.
Stability Analysis	input-to-state stability, Strict dissipativity	Exponential	Nominal	Strict dissipativity	Practical	Convergence	Different stability guarantees.

(LMIs). The LMIs can be efficiently solved using convex optimization techniques. This method ensures that both stability and optimal economic performance are achieved.

A comprehensive comparison between the proposed approach and existing methods is presented in [Table 1](#), highlighting key features across several criteria.

We would like to emphasize that, to the best of our knowledge, no previous work has addressed Robust Tube-based Economic MPC within an LPV framework. This constitutes the main novelty of our contribution.

Building upon this novelty, the main contributions of the present work are summarized as follows:

- Robust Tube-Based Economic MPC for Nonlinear Systems:** A tube-based REMPC scheme that handles UBB disturbances in nonlinear plants via LPV modeling.
- LPV-based Modeling and Efficient QP Solving:** Nonlinear dynamics are captured by scheduling-variable LPV blends (no linearization) enabling fast quadratic programming.
- Robust Tube Design with Zonotope-Bounded Uncertainty:** Parameter-dependent gains and zonotopic reachable sets jointly tighten constraints and bound error propagation.
- LMI-based Terminal Setup and Stability Theory:** Novel terminal controller and region LMIs guarantee recursive feasibility and input-to-state stability.
- Robust Control Design:** The MPC formulation ensures robust performance and stability by employing an $LPV - H_\infty$ gain. This control gain is used to tighten constraints over the prediction horizon, considering the potential impacts of disturbances. Importantly, this gain does not need to match the gain of the terminal controller, allowing flexibility in tuning to expand the domain of attraction.

This study presents two illustrative examples to demonstrate the efficacy of the proposed method. The first example focuses on a control problem applied to a nonlinear Continuous Stirred Tank Reactor (CSTR) [20], showcasing the method's capability to handle nonlinearities and achieve desired performance in chemical process control. The second, and more critical, example involves a robust economic MPC based planner for an electric autonomous vehicle in a racing scenario [14]. This example highlights the practical application and robustness of our method in dynamically complex and high-stakes environments, ensuring optimal performance and safety under varying conditions. The REMPC approach effectively integrates economic objectives with system dynamics, providing a comprehensive solution for controlling nonlinear systems subject to parameter variations and uncertainties.

The rest of the paper is structured as follows: [Section 2](#) presents the problem formulation, detailing the theoretical framework and the key concepts underpinning our study. [Section 3](#) delves into stability analysis and design, providing an in-depth examination of the methodologies and techniques employed. [Section 4](#) offers a comprehensive case study, featuring two examples: a CSTR and, most importantly, an autonomous racing vehicle, which serves as the focal point of our investigation. Finally, [Section 5](#) concludes the paper by summarizing the findings, discussing their implications, and suggesting potential directions for future research.

2. Problem statement

2.1. LPV formulation

Our objective is to control a discrete time disturbance affected nonlinear system represented by the following state space realization:

$$x(k+1) = f(x(k), u(k), w(k)), \quad (1)$$

where $x \in \mathbb{R}^{n_x}$ representing the state vector, $u \in \mathbb{R}^{n_u}$ is the control input vector, while $f(x(k), u(k), w(k))$ is a nonlinear function describing system dynamics after modelling and it assumed to be of a class C^1 , the term $w \in \mathbb{R}^{n_w}$ is considered as additive disturbance to

encapsulates all unmodelled dynamics and exogenous disturbances affecting the plant. It is constrained within a compact set centered at the origin, such that $\|w\| \leq \bar{w}$. This condition implies that w belongs to $\mathcal{W} \subseteq \mathbb{R}^{n_w}$, where $\mathcal{W}$ represents the set of disturbances.

State and control constraints apply to the system like so,

$$(x(k), u(k)) \in \mathbb{Z}, k \in \mathbb{I}_{\geq 0}, \quad (2)$$

where $\mathbb{Z} := \{(x, u) \in \mathbb{R}^{n_x \times n_u}, x \in \mathbb{R}^{n_x}, u \in \mathbb{R}^{n_u} | a_j x + b_j u \leq h_j, \forall j \in \mathbb{I}_{[1, n_r]}\}$, where h_j is a known bound and a_j and b_j are state and input constraints respectively.

It is important to note that

$$x \in \mathcal{X} \subseteq \mathbb{R}^{n_x}, u \in \mathcal{U} \subseteq \mathbb{R}^{n_u}, \quad (3)$$

where $\mathcal{X}$ and $\mathcal{U}$ are defined as:

$$\mathcal{X} := \left\{ x \in \mathbb{R}^{n_x} \mid H_x x \leq h_x \right\} \quad (4a)$$

$$\mathcal{U} := \left\{ u \in \mathbb{R}^{n_u} \mid H_u u \leq h_u \right\} \quad (4b)$$

Remark 1. In the context of application scenarios, the additive disturbance at time step k , denoted as $w(k)$, can be decomposed into two distinct components:

$$w(k) = w_d(k) + w_r(k). \quad (5)$$

The term $w_d(k)$ includes all possible external disturbances that may impact the system. Concurrently, $w_r(k)$ accounts for the residual term arising from imperfect predictions of the scheduling parameters along the prediction horizon. This decomposition allows for a more nuanced understanding of the sources contributing to the overall disturbance, enabling a more robust analysis and mitigation strategy within the application framework.

The motivation of our work is that instead of directly using the nonlinear systems to tackle the robust tube economic model predictive control, we will use an equivalent and thus effective representation.

Moving forward, our immediate objective is to determine a function $\phi(x(k), u(k))$ that effectively maps endogenous scheduling variables. This mapping should ensure that the nonlinear model given by Eq. (1) can be precisely rewritten into what is commonly referred to as the quasi-Linear Parameter Varying (quasi-LPV) formulation. This transformation, facilitated by the nonlinear embedding approach mentioned in [21,22], yields the representation:

$$x(k+1) = A(\theta(k))x(k) + B(\theta(k))u(k) + w(k) \quad (6)$$

where $A(\theta(k)) \in \mathbb{R}^{n_x \times n_x}$ and $B(\theta(k)) \in \mathbb{R}^{n_x \times n_u}$ are parameter-dependent matrices representing the system and the input matrices, respectively. They are contingent on the scheduling vector $\theta(k) := [\theta_1(k), \theta_2(k), \dots, \theta_{n_\theta}(k)] \in \mathbb{R}^{n_\theta}$, where n_θ is the number of scheduling variables.

note 1. The reader is referred to [21] and [22] for a detailed discussion of the transformation procedure and its applicability to nonlinear systems.

The LPV model (6) is represented in a polytopic LPV fashion, such that the vector θ belongs to a convex polytope

$$\Theta := \{\theta(k) \in \mathbb{R}^{n_\theta} | \theta(k) = \sum_{j=1}^N \mu_j(\theta(k)) \theta^j, \sum_{j=1}^N \mu_j(\theta(k)) = 1, \mu_j(\theta(k)) \geq 0\} \quad (7)$$

defined by N vertices, where θ^j are the vertices of the polytope and $\mu_j(\theta(k))$ are polytopic interpolation functions. Hence, as $\theta(k)$ varies inside the convex polytope Θ , the matrices of the system (6) vary inside a corresponding polytope Π , which is defined by the convex hull (Co) of N local matrix vertices $[A_j, B_j]$, $j \in [1, \dots, N]$, that is,

$$\Pi := Co\{[A_1 B_1], [A_2 B_2], \dots, [A_N B_N]\}, \quad (8)$$

On the other hand, the system matrices in (6) are expressed as

$$A(\theta(k)) = \sum_{n=1}^N \mu_j(k) A_j, \quad B(\theta(k)) = \sum_{n=1}^N \mu_j(k) B_j, \quad (9)$$

Assumption 1. Under this assumption, we posit that $\theta(k) = \phi(x(k), u(k)) \in \mathbb{R}^{n_\theta}$, $k \geq 0$, and $\theta(k)$ resides within a compact set Θ , and the function $\phi(x, u)$ serves as the scheduling function that embeds the nonlinearities arising from the time-varying parameters, which are functions of the system states and control inputs.

Assumption 2. For all $\theta \in \Theta$, the pair $(A(\theta), B(\theta))$ is structurally stabilizable.

Remark 2. It is worth noting that the system states $x(k)$ are continuously measured online. Additionally, since $\phi(\cdot)$ is a predetermined function, the scheduling parameters are readily available without the need for direct measurement; instead, they are computed from the measured state information. However, it is important to emphasize that the future scheduling trajectory, denoted by the future values $\theta(k+j)$ for all $j \in \mathbb{I}_{[1, H_p]}$, remains unknown.

However, due to unknown disturbances in the system dynamics, precise prediction of future system states becomes infeasible. Nevertheless, a prevailing strategy in robust MPC involves identifying an invariant set characterizing the deviation between the actual system and the nominal system. At this end we define a nominal system by setting $w \equiv 0$ in system (1)

$$\bar{x}(k+1) = f(\bar{x}(k), \bar{u}(k)) \quad (10)$$

$$\bar{x} \in \bar{\mathcal{X}}, \bar{u} \in \bar{\mathcal{U}} \quad (11)$$

where $\bar{\mathcal{X}} \subseteq \mathcal{R}^{n_x}$, $\bar{\mathcal{U}} \subseteq \mathcal{R}^{n_u}$ are the state and input sets for the nominal system respectively.

We consider the deviation between the disturbed system (1) and the disturbance-free (nominal) system (10) is defined as $e(k) = x(k) - \bar{x}(k)$. The error dynamics are then given by

$$e(k+1) = f(x(k), u(k), w(k)) - f(\bar{x}(k), \bar{u}(k)) \quad (12)$$

The accompanying nominal system of the LPV system (6):

$$\bar{x}(k+1) = A(\theta(k))\bar{x}(k) + B(\theta(k))\bar{u}(k) \quad (13)$$

$$\bar{x} \in \bar{\mathcal{X}}, \bar{u} \in \bar{\mathcal{U}} \quad (14)$$

where $\bar{\mathcal{X}} \subseteq \mathcal{R}^{n_x}$, $\bar{\mathcal{U}} \subseteq \mathcal{R}^{n_u}$ are the state and input sets for the nominal system respectively.

Remark 3. The computation of the sets $\bar{\mathcal{X}}$ and $\bar{\mathcal{U}}$ will be discussed later in the paper.

At this stage, the new error dynamics become

$$e(k+1) = A(\theta(k))e(k) + B(\theta(k))(u(k) - \bar{u}(k)) + w(k) \quad (15)$$

Defining the control signal to be like:

$$u(k) = \bar{u}(k) + K(\theta(k))(x(k) - \bar{x}(k)) \quad (16)$$

such that $\bar{u}(k) = u^*(0|k)$ is the solution of the OCP (43). And hence:

$$e(k+1) = (A(\theta(k)) + B(\theta(k))K(\theta(k)))e(k) + w(k) \quad (17)$$

where $K(\theta(k))$ is a state feedback gain chosen such that $(A(\theta(k)) + B(\theta(k))K(\theta(k)))$ is stable. Where this local controller is used to compensate the mismatch between the nominal LPV model (13) and the model affected by additive external disturbances (6) and used to refine the zonotopic tubes of the robust MPC later in the paper.

Moreover, the system outputs can be defined as follows

$$y(k) = Cx(k) + Du(k) \quad (18)$$

2.2. Bounding uncertainty effect with zonotopes

In this section, our attention is directed towards the computation of reachable sets, notably identified as the calculation of one-step forward-reachable sets. The methodology employed here leverages a zonotopic-based representation, which proves to be a powerful tool in this context. The primary objective of applying reachability theory in this scenario encompasses two fundamental aspects. Firstly, it involves the establishment of bounds on the maximum achievable values for the mismatch error (17). This error denotes the mismatch between the prediction model and the actual measurements at each sampling time. Secondly, the application of reachability theory aims to refine the tube of trajectories along the prediction horizon within the framework of the Model Predictive Control (MPC) problem.

Definition 1 ([19]). A set $\mathcal{Z}$ is a *zonotope* if there exists the pair $(H_z, z_0) \in \mathbb{R}^{n \times r} \times \mathbb{R}^n$ such that

$$\mathcal{Z} = \{ z_0 + H_z \zeta : \zeta \in \mathbb{R}^r, \|\zeta\|_\infty \leq 1 \}, \quad (19)$$

where z_0 is the center and H_z is the generator matrix.

Definition 2 ([8] Robust Positive Invariant (RPI) set). Given the system $x(k+1) = (A(\theta(k)) + B(\theta(k))K(\theta(k)))x(k) + w(k)$, where $(A(\theta(k)) + B(\theta(k))K(\theta(k)))$ is stable and $w(k) \in \mathcal{W}, \forall k \in H_p$. A set is said to be an RPI set of the close loop system if

$$(A(\theta(k)) + B(\theta(k))K(\theta(k)))\mathcal{Z} \oplus \mathcal{W} \subseteq \mathcal{Z} \quad (20)$$

By Definition 2, the RPI set can be explicitly formulated as $\mathcal{Z}_\infty = \bigoplus_{i=0}^\infty (A(\theta(k)) + B(\theta(k))K(\theta(k)))^i \mathcal{W}$, where $\oplus$ denotes the Minkowski sum. Considering the set $\mathcal{W}$ defined by

$$\mathcal{W} := \{ w \in \mathcal{R}^{n_w} \mid \|w\| \leq \bar{w} \} \quad (21)$$

It is important to mention that it is impossible to obtain $\mathcal{Z}_\infty$ owing to the infinity of iterations. Consequently, to address this challenge, an outer approximation method with a given precision is utilized to compute the sequence $\mathcal{Z}_i$ such that $i \in \mathbb{N}$. The resulting set $\mathcal{Z}_i$ is used to replace the RPI set $\mathcal{Z}_\infty$ [23].

In this study, zonotopes are employed for computing reachable sets due to their simplicity in set operations. The one-step forward-reachable set, described by Eq. (20), can be reformulated as:

$$\mathcal{Z}(i+1) = ((A(\theta(k+i)) + B(\theta(k+i))K(\theta(k+i))) \circ \mathcal{Z}(i)) \oplus \mathcal{W}(i), \quad i \in [0, H_p]. \quad (22)$$

Considering Eq. (22) and the initial sets $\mathcal{X}$ and $\mathcal{U}$ defined as boxes with state and control input constraints, at each time iteration k , $H_p + 1$ zonotopic reachable sets are systematically computed. The initial set $\mathcal{Z}(0)$ is considered as $\mathcal{W}$. The computation of each reachable set $\mathcal{Z}(i+1)$ depends on its past realization $\mathcal{Z}(i)$, the scheduling vector $\theta(i)$, system matrices $A(\theta(i))$ and $B(\theta(i))$, controller $K(\theta(i))$, and uncertainty/disturbance set $\mathcal{W}$. Then, utilizing tightened bounds, if $(x(k+i|k), \bar{u}(k+i)) \in \mathbb{Z} \ominus \mathcal{Z}(i)$ and $x(k+i) - \bar{x}(k+i|k) \in \mathcal{Z}(i)$, it follows that $x(k+i+1) - \bar{x}(k+i+1|k) \in \mathcal{Z}(i+1)$.

For all $i \in [0, H_p]$, representing the one-step-ahead deviation from the nominal state prediction $\bar{x}(k+i|k)$ and the real state value $x(k+i)$ being bounded synthetically as $\bar{x}(k+i|k+1) \in \bar{x}(k+i|k) \oplus \mathcal{Z}(i)$, for any admissible sequence of inputs and scheduling parameter predictions. These one-step-ahead disturbance propagation sets $\mathcal{Z}(i)$, thus, bound the difference between the predictions made in k and $k+1$, and therefore can be used to guarantee recursive feasibility and constraint satisfaction of the MPC based on nominal predictions.

Finally, these zonotopic reachable sets are used for computing refined state and input sets along the prediction horizon H_p at each time k , known as tubes, through the Pontryagin difference $\ominus$ as follows:

$$\begin{aligned} \bar{\mathcal{X}}(k+i) &= \mathcal{X} \ominus \mathcal{Z}(k+i), \quad i \in [0, H_p] \\ \bar{\mathcal{U}}(k+i) &= \mathcal{U} \ominus K(\theta(k+i))\mathcal{Z}(k+i), \quad i \in [0, H_p - 1]. \end{aligned} \quad (23)$$

Assumption 3. The tightened sets $\bar{\mathcal{X}}$ and $\bar{\mathcal{U}}$ are assumed non empty.

Since there are mismatches between the real system and the nominal model, the predicted evolution using the nominal model might differ from the real evolution of the system. In order to consider this effect in the controller synthesis, a bound on the difference between the predicted and the real evolution is given in the following theorem.

Lemma 1 ([24]). *The bounding box of a zonotope can be obtained using its interval hull.*

Theorem 1 ([25]). *Consider a Linear Parameter-Varying (LPV) system described in (6), with real state variables $x(k+i+1) \in \mathbb{R}^{n_x}$, and nominal predicted state variables $\bar{x}(k+i+1|k) \in \mathbb{R}^{n_x}$, the control input is defined as $u(k) = \bar{u}(k+i) + K(\theta(k+i))\Delta(k+i) \in \mathbb{R}^{n_u}$, and scheduling parameters $\theta(k+i), \theta(k+i|k) \in \mathbb{R}^{n_\theta}$ for all $i \in [0, H_p]$. Define the error term as $\Delta(k+i|k) = x(k+i) - \bar{x}(k+i|k)$.*

The error dynamics equation is given by:

$$\Delta(k+i+1|k) = A_K(\theta(k+i))\Delta(k+i|k) + w_r(k+i) + w_d(k+i), \quad (24)$$

where $w_r(k+i)$ represents residual terms resulting from scheduling parameter estimates, and $w_d(k+i)$ represents disturbance terms.

Using zonotopic notation, the propagated error is contained within a zonotope using reachable sets:

$$\Delta(k+i+1|k) \in \mathcal{Z}(i) = (A_K(\theta(k+i))^i \mathcal{W}(i)), \quad i = 0, 1, \dots, H_p, \quad (25)$$

where $\mathcal{W}(i) = [\mathcal{W}_d(i) \quad \mathcal{W}_r(i)]$, and $\mathcal{W}_r(i)$ is defined as:

$$\mathcal{W}_r(i) = [\Delta_A(i) \quad \Delta_B(i)]Z, \quad (26)$$

with Z being an interval matrix containing the admissibility bounds of the LPV system.

The propagation of uncertainty is quantified by a function σ using the bounding box, which is defined by the interval hull of the zonotope, as described in [24]. We consider σ as a class $\mathcal{K}$ function:

$$\|\Delta(k+i|k)\|_1 \leq \sigma^i(\|w(k)\|), \quad \forall i \in [0, H_p]. \quad (27)$$

Proof. Consider the state and input variables $x(k+i+1), \bar{x}(k+i+1|k) \in \mathbb{R}^{n_x}$, $\bar{u}(k+i) \in \mathbb{R}^{n_u}$, and scheduling parameters $\theta(k+i), \theta(k+i|k) \in \mathbb{R}^{n_\theta}$, for all $i \in [0, H_p]$. Let the error term be $\Delta(k+i|k) = x(k+i) - \bar{x}(k+i|k)$.

We derive the error dynamics as follows:

$$\begin{aligned} \Delta(k+i+1|k) &= A(\theta(k+i))x(k+i) + B(\theta(k+i))u(k+i) + w_d(k+i) \\ &\quad - (A(\theta(k+i|k))\bar{x}(k+i|k) + B(\theta(k+i|k))\bar{u}(k+i)), \\ &= (A(\theta(k+i)) + B(\theta(k+i))K(\theta(k+i)))\Delta(k+i|k) \\ &\quad + \Delta_A(i)\bar{x}(k+i|k) + \Delta_B(i)\bar{u}(k+i) + w_d(k+i), \\ &= A_K(\theta(k+i))\Delta(k+i|k) + w_r(k+i) + w_d(k+i), \end{aligned} \quad (28)$$

$$\Delta(k+i+1|k) = A_K(\theta(k+i))\Delta(k+i|k) + w(k+i).$$

Using zonotopic notation, the propagated error is contained within the zonotope:

$$\Delta(k+i+1|k) \in \mathcal{Z}(i) = (A_K(\theta(k+i))^i \mathcal{W}(i)), \quad i = 0, 1, \dots, H_p, \quad (29)$$

where $\mathcal{W}(i) = [\mathcal{W}_d(i) \quad \mathcal{W}_r(i)]$, and $\mathcal{W}_r(i)$ is defined as:

$$\mathcal{W}_r(i) = [\Delta_A(i) \quad \Delta_B(i)]Z, \quad (30)$$

with Z being an interval matrix containing the admissibility bounds of the LPV system.

Explicitly, the set $\mathcal{Z}(i)$ can be written as:

$$\mathcal{Z}(i) = \mathcal{W} \oplus A_K(\theta(k+i))\mathcal{W} \oplus \cdots \oplus A_K^i(\theta(k+i))\mathcal{W}, \quad i = 0, 1, \dots, H_p. \quad (31)$$

According to Lemma 1, the disturbance propagation is bounded by the interval hull of the zonotope:

$$\|\Delta(k+i|k)\|_1 = \|\mathcal{Z}(i)\|_1, \quad i = 0, 1, \dots, H_p. \quad (32)$$

Denoting the matrix $M(i)$ as:

$$M(i) = I \oplus A_K(\theta(k+i)) \oplus \cdots \oplus A_K^{i-1}(\theta(k+i)), \quad (33)$$

we have:

$$\|\mathcal{Z}(i)\|_1 = \|M(i)\mathcal{W}\|_1. \quad (34)$$

Using the sub-multiplicative property of the norm, we obtain:

$$\|\mathcal{Z}(i)\|_1 \leq \|M(i)\|_1 \cdot \|\mathcal{W}\|_1. \quad (35)$$

Therefore, the uncertainty propagation is bounded by:

$$\|\Delta(k+i|k)\|_1 \leq \|\mathcal{Z}(i)\|_1 \leq \|M(i)\|_1 \cdot \|\mathcal{W}\|_1 = \sigma^i(\|w(k)\|), \quad i = 0, 1, \dots, H_p. \quad (36)$$

Finally, the propagation of uncertainty is quantified by a function σ as:

$$\|\Delta(k+i|k)\|_1 \leq \sigma^i(\|w(k)\|), \quad i = 0, 1, \dots, H_p. \quad (37)$$

Since σ satisfies continuity, strict monotonicity, and $\sigma(0) = 0$ (as reasoned in the context of bounding functions in zonotopic analysis), we consider σ to be a class $\mathcal{K}$ function, which formalizes the bounding of uncertainty propagation. This ends the proof. $\square$

2.3. The novel robust economic model predictive control

Economic Model Predictive Control (EMPC) aims to minimize economic costs through control functions, addressing factors like production costs, energy savings, and efficiency. In this paper, we introduce a novel robust EMPC framework for nonlinear disturbed systems using the qLPV approach. Drawing inspiration from the LPV-based EMPC for nonlinear nominal systems suggested in [5], our approach stands out by its handling of disturbances through the tube-based method. By employing qLPV nominal predictions, our method operates much faster compared to existing techniques, aligning the numerical complexity more closely with that of a QP rather than an NLP. Furthermore, we integrate constraint tightening, terminal cost, and terminal constraints to ensure the recursive feasibility and stability of the resulting closed-loop system.

2.3.1. Motivation: Robust economic nonlinear model predictive control

Firstly, we present the formulation of the nonlinear Optimal Control Problem (OCP) within the framework of Robust Economic Nonlinear Model Predictive Control (RENMP) as follows:

$$\min_{\bar{u}(k), \bar{x}(k)} L_e(\bar{x}(k), \bar{u}(k)) = \sum_{i=0}^{H_p-1} \ell(\bar{x}(i), \bar{u}(i)), \quad (38a)$$

subject to

$$\bar{x}(i+1|k) = f(\bar{x}(i|k), \bar{u}(i|k)), \quad i \in [0, H_p - 1] \quad (38b)$$

$$\bar{x}(i|k) \in \mathcal{X} \ominus \mathcal{Z}(i|k), \quad i \in [0, H_p - 1] \quad (38c)$$

$$\bar{u}(i|k) \in \mathcal{U} \ominus K(\theta(i|k))\mathcal{Z}(i|k), \quad i \in [0, H_p - 1] \quad (38d)$$

$$\bar{x}(0|k) = x(k) \quad (38e)$$

$$\bar{x}(H_p) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p|k). \quad (38f)$$

where $\ell(\bar{x}, \bar{u}) : \bar{\mathcal{X}} \times \bar{\mathcal{U}} \rightarrow \mathcal{R}$ represents an economic stage cost, noting that the set $\mathcal{X}_f \ominus \mathcal{Z}(H_p|k) \subseteq \mathcal{X}_f$ will contain $\bar{x}_s^*$ in its interior. Here, $\bar{x}$ and $\bar{u}$ are sequences of states and control inputs over a prediction horizon H_p :

$$\bar{u}^*(k) = [\bar{u}^*(0|k), \bar{u}^*(1|k), \dots, \bar{u}^*(H_p-1|k)]^T,$$

$$\bar{x}^*(k) = [\bar{x}^*(0|k), \bar{x}^*(1|k), \dots, \bar{x}^*(H_p|k)]^T.$$

This controller aims to steer the system towards the nominal optimal reachable steady state. This state is determined using the implicit form of the optimization problem introduced in Definition (39).

Definition 3. The nominal optimal reachable steady state and input, $(\bar{x}_s^*, \bar{u}_s^*)$, of the nonlinear system (10) satisfy

$$(\bar{x}_s^*, \bar{u}_s^*) = \arg \min_{(\bar{x}, \bar{u})} \ell_e(\bar{x}, \bar{u}) \quad (39a)$$

subject to

$$\bar{x} = f(\bar{x}, \bar{u}) \quad (39b)$$

$$\bar{x} \in \bar{\mathcal{X}} \quad (39c)$$

$$\bar{u} \in \bar{\mathcal{U}} \quad (39d)$$

It is important to highlight that in standard tracking MPC, $\ell_e(\bar{x}, \bar{u})$ is usually constructed to be positive definite relative to $(\bar{x}_s^*, \bar{u}_s^*)$, ensuring closed-loop stability. Conversely, EMPC formulates $\ell_e(\bar{x}, \bar{u})$ by considering economic criteria, which might not ensure positive definiteness. As a result, standard MPC stability analysis does not guarantee stability or convergence in optimal economic operation.

Although efficient nonlinear optimization algorithms have been developed in recent years, optimization problems with nonlinear constraints tend to be computationally more expensive than those related to a linear MPC based on linear/QP. Moreover, nonlinear programming cannot guarantee that the solution of the nonlinear optimization problem yields the global optimum due to the lack of convexity. However, despite these challenges, strategies such as linearization, approximation, and the adoption of linear parameter varying (LPV) formulation are crucial for achieving reliable real-time solutions.

2.3.2. Contribution: The proposed robust economic MPC based on LPV formulation

In opposition to problem (38), we propose herein a novel robust formulation, which makes use of the LPV model and removes the nonlinear constraints from the prior. Ensuring robust performance, we employ a local LPV control law and Robust positive invariant (RPI) sets to tighten the constraints associated with system states and control inputs. Employing the robust tube-based methodology, the refined constraints, represented as zonotopes, are used in the formulation of the robust economic model predictive controller.

In this paper, the nonlinear optimal control problem (OCP) described by equation (38) is reformulated into the problem outlined in Definition 4

Definition 4. At each time instant k , the following optimization problem should be solved considering that the state x and the vector of parameters θ are known at the beginning of H_p :

$$\min_{\bar{u}(k) \in \bar{\mathcal{U}}(k)} L_e(x(k), \bar{u}(k)) \quad (40a)$$

subject to

$$\bar{x}(i+1|k) = A(\theta(i|k))\bar{x}(i|k) + B(\theta(i|k))\bar{u}(i|k), \quad i \in [0, H_p - 1] \quad (40b)$$

$$\theta(i+1|k) = \phi(x(k), u(k)), \quad (40c)$$

$$\theta(0|k) = \theta(k), \quad (40d)$$

$$\bar{x}(0|k) = x(k), \quad (40e)$$

$$\bar{x}(i|k) \in \mathcal{X} \ominus \mathcal{Z}(i|k), \quad i \in [0, H_p - 1] \quad (40f)$$

$$\bar{u}(i|k) \in \mathcal{U} \ominus K(\theta(i|k))\mathcal{Z}(i|k), \quad i \in [0, H_p - 1] \quad (40g)$$

$$\bar{x}(H_p) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p|k). \quad (40h)$$

where

$$\bar{\mathbf{u}}^*(k) = [\bar{u}^*(0|k), \bar{u}^*(1|k) \dots \bar{u}^*(H_p - 1|k)]^T \quad (40i)$$

$$\bar{\mathbf{x}}^*(k) = [\bar{x}^*(0|k), \bar{x}^*(1|k) \dots \bar{x}^*(H_p|k)]^T \quad (40j)$$

are the optimal solutions of the robust economic MPC.

2.3.3. Quadratic programming (QP) formulation:

It is evident that even with the introduction of the LPV model into MPC, the optimization problem remains nonlinear due to the dependence of the time-varying parameters on control input and state sequences, which are unknown and must be determined a priori. However, various methods exist for estimating the scheduling parameters along the prediction horizon, such as those discussed in [25,26] where they use a recursive procedure based on simple Taylor expansions. Hence, we adopt the approach used in [27], which uses optimal state and input sequences from previous iteration and shifting them in time to facilitate the estimation of $x(i|k)$ and $u(i|k)$ along the prediction horizon H_p , which can be expressed as $\hat{\theta}(i|k) = \phi(\hat{x}(i|k), \hat{u}(i|k))$.

Utilizing this technique, the LPV-based MPC problem is transformed into a linear time-varying (LTV) one by approximating θ with its estimation, $\hat{\theta}$, across H_p . This conversion is facilitated by defining a sequence of estimated time-varying parameters $\Theta(k)$, containing the estimated parameters along H_p .

$$\Theta(k) = [\hat{\theta}(0|k), \hat{\theta}(1|k), \dots, \hat{\theta}(H_p - 1|k)]^T \in \mathbb{R}^{H_p n_\theta}. \quad (41)$$

with $\Theta(k)$, the sequence of predicted states in terms of the sequence of control inputs can be expressed as

$$\bar{\mathbf{x}}(k) = A(\Theta(k))x(k) + B(\Theta(k))\bar{\mathbf{u}}(k), \quad (42)$$

where $A(\Theta(k))$ and $B(\Theta(k))$ are matrices encoding the dynamics with respect to the estimated parameters.

Hence, using (42) the REMPC optimization problem (40) can be rewritten as

$$\min_{\bar{\mathbf{u}}(k) : \bar{\mathbf{x}}(k)} L_e(A(\Theta(k))\bar{\mathbf{x}}(k) + B(\Theta(k))\bar{\mathbf{u}}(k)) \quad (43a)$$

subject to

$$\theta(0|k) = \theta(k), \quad (43b)$$

$$\bar{\mathbf{x}}(0|k) = \mathbf{x}(k), \quad (43c)$$

$$\bar{\mathbf{x}}(i|k) \in \mathcal{X} \ominus \mathcal{Z}(i|k), \quad i \in [0, H_p - 1] \quad (43d)$$

$$\bar{\mathbf{u}}(i|k) \in \mathcal{U} \ominus K(\theta(i|k))\mathcal{Z}(i|k), \quad i \in [0, H_p - 1] \quad (43e)$$

$$\bar{\mathbf{x}}(H_p) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p|k). \quad (43f)$$

with

$$A(\Theta(k)) = \begin{pmatrix} A(\hat{\theta}(k)) \\ A(\hat{\theta}(k+1))A(\hat{\theta}(k)) \\ \vdots \\ A(\hat{\theta}(k+H_p-1))A(\hat{\theta}(k+H_p-2)) \dots A(\hat{\theta}(k)) \end{pmatrix} \quad (44)$$

and

$$B(\Theta(k)) = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ B(\hat{\theta}(k)) & 0 & 0 & \dots & 0 \\ A(\hat{\theta}(k+1))B(\hat{\theta}(k)) & B(\hat{\theta}(k+1)) & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ A(\hat{\theta}(k+H_p-1)) \dots A(\hat{\theta}(k+1))B(\hat{\theta}(k)) & A(\hat{\theta}(k+H_p-1)) \dots A(\hat{\theta}(k+2))B(\hat{\theta}(k+1)) & B(\hat{\theta}(k+H_p-1)) & \dots & 0 \end{pmatrix} \quad (45)$$

By employing this formulation, the REMPC optimization problem can be rewritten as a quadratic programming (QP) problem, which is notably more computationally efficient than tackling a nonlinear optimization problem directly. This is achieved by parameter-varying estimation at each discrete time step k using the approach described earlier:

Initially, at time step $k = 0$, the optimization problem (43) is solved as a linear problem by substituting the LPV model (40c) with an LTI model. This LTI model is constructed by assuming a constant θ over the horizon H_p , i.e., $\theta(0|k) = \theta(1|k) = \theta(2|k) = \dots = \theta(H_p - 1|k)$, where $\theta(0|k) = \phi(\mathbf{x}(0), \mathbf{u}(0))$ and $\mathbf{u}(0) = 0$.

Subsequently, the sequence of estimated time-varying parameters $\Theta(k)$ in (41) is computed using $\hat{\theta}(i|k) = \hat{\phi}(\hat{\mathbf{x}}(i|k), \hat{\mathbf{u}}(i|k))$. Here, the estimated states and control inputs are derived from the optimal state and control input sequences, $\bar{\mathbf{x}}^*(\mathbf{k} - 1)$ and $\bar{\mathbf{u}}^*(\mathbf{k} - 1)$, which are obtained by solving (43) in the previous iteration and appropriately shifted in time.

Tightened constraints are introduced to the optimization problem to account for potential disturbances and uncertainties. These constraints ensure that the predicted state and input trajectories remain within feasible regions.

The tightened state constraint is denoted by $\mathcal{X} \ominus \mathcal{Z}(i|k)$, where $\mathcal{X}$ represents the original state constraint and $\mathcal{Z}(i|k)$ represents the tube around the predicted state at time i . Subtracting this tube shrinks the feasible state space, guaranteeing that the predicted state $\bar{\mathbf{x}}(i|k)$ lies within a region that accounts for potential deviations from the nominal prediction.

Similarly, the tightened input constraint is expressed as $\mathcal{U} \ominus K(\theta(i|k))\mathcal{Z}(i|k)$. Here, $\mathcal{U}$ denotes the original input constraint, $K(\theta(i|k))$ represents the gain matrix associated with the time-varying parameters $\theta(i|k)$, and $\mathcal{Z}(i|k)$ again represents the tube around the predicted state at time i . This constraint considers the influence of time-varying parameters and potential disturbances on the feasible input space, ensuring that the predicted input $\bar{\mathbf{u}}(i|k)$ remains within acceptable bounds.

The terminal constraint is specified as $\bar{\mathbf{x}}(H_p) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p|k)$, where $\mathcal{X}_f$ represents the terminal set, H_p is the prediction horizon, and $\mathcal{Z}(H_p|k)$ represents the tube around the predicted terminal state. Subtracting the terminal tube from the terminal set confines the predicted terminal state $\bar{\mathbf{x}}(H_p)$ to a region that accounts for uncertainties at the end of the prediction horizon. Notably, the tightened terminal set, $\mathcal{X}_f \ominus \mathcal{Z}(H_p|k)$, guarantees that the nominal optimal reachable steady state $\bar{\mathbf{x}}_s^*$ remains within its interior. This ensures that the predicted terminal state converges towards the desired steady state while incorporating the effects of uncertainties.

3. Stability analysis and design

Given disparities between the real system and the nominal model, the anticipated progression using the nominal model may deviate from the actual evolution of the system. To account for this discrepancy in controller synthesis, a constraint is provided on the disparity between the projected and actual evolution in the following lemma.

Definition 5. A function $\gamma : \bar{\mathcal{X}} \rightarrow \mathbb{R}$ can be found such that the nominal system (13) exhibits dissipative behavior with respect to the stock rate $s(\bar{\mathbf{x}}, \bar{\mathbf{u}})$ if

$$\gamma(A(\theta(k))\bar{\mathbf{x}}(k) + B(\theta(k))\bar{\mathbf{u}}(k)) - \gamma(\bar{\mathbf{x}}) \leq s(\bar{\mathbf{x}}, \bar{\mathbf{u}}), \quad (46)$$

for all $(\bar{x}, \bar{u}) \in (\bar{\mathcal{X}} \times \bar{\mathcal{U}})$. Additionally, if $\rho : \bar{\mathcal{X}} \rightarrow \mathbb{R}_{\geq 0}$ is strictly positive, then

$$\gamma(A(\theta(k))\bar{x}(k) + B(\theta(k))\bar{u}(k)) - \gamma(\bar{x}) \leq -\rho(\bar{x} - \bar{x}_s^*) + s(\bar{x}, \bar{u}), \quad (47)$$

and it can be stated that the system demonstrates strict dissipative behavior.

Assumption 4 ([5]). The nominal system (13) is assumed to exhibit strict dissipativity regarding the stock rate $s(\bar{x}, \bar{u})$, which is defined as $s(\bar{x}, \bar{u}) = \ell(\bar{x}, \bar{u}) + \ell(\bar{x}_s^*, \bar{u}_s^*)$.

Assumption 5 ([5]). Both the stage cost function ℓ and the nominal model (13) are assumed to be continuous over the domain $(\bar{\mathcal{X}}, \bar{\mathcal{U}})$. Additionally, the terminal cost function $V_f(\cdot)$ is continuous within the set $\mathcal{X}_f$.

Assumption 6 ([5]). It is assumed that the storage function $\gamma(\bar{x})$ remains continuous within the domain $(\bar{\mathcal{X}}, \bar{\mathcal{U}})$.

Assumption 7 (Stability assumption [5]). Assuming a compact terminal region $\mathcal{X}_f \ominus \mathcal{Z}(H_p)$ that includes the point $\bar{x}_s^*$ within its interior, there exists a control law $K_f(\theta(k))\bar{x}(k) : \mathcal{X}_f \ominus \mathcal{Z}(H_p) \rightarrow \bar{\mathcal{U}}$ satisfying the following condition:

$$V_f((A(\theta(k)) + B(\theta(k))K_f(\theta(k))\bar{x}(k)) - V_f(x) \leq -\ell(\bar{x}_s^*, K_f(\theta(k))\bar{x}) + \ell(\bar{x}_s^*, \bar{u}_s^*) \quad (48)$$

Remark 4. Implication of Assumption 7 suggests that the set $\mathcal{X}_f \ominus \mathcal{Z}(H_p)$ remains invariant for all $k \geq 0$ under the control law defined as $\bar{u}(\theta(k)) = K_f(\theta(k))\bar{x}(k)$.

Remark 5. It's reasonable to assume, without loss of generality, that $V_f(\bar{x}_s^*) = 0$ for all $k \geq 0$, as Assumption 7 is the only condition imposed on V_f . It's important to highlight that unlike in the standard MPC problem, $V_f(\bar{x})$ may not necessarily demonstrate positive definiteness concerning $\bar{x}_s^*$.

Assumption 8. It is assumed that the feedback gain $K(\theta(k))$ stabilizes the error system (17), ensuring that $A(\theta(k)) + B(\theta(k))K(\theta(k))$ is stable.

Assumption 9 ([15]). The compact convex set $\mathcal{X}_f$ is subject to the following conditions:

$$(A(\theta(H_p - 1)) + B(\theta(H_p - 1))K_f(\theta(H_p - 1)))\mathcal{X}_f \oplus \mathcal{Z}(H_p) \subseteq \mathcal{X}_f, \quad (49)$$

$$\mathcal{X}_f \subseteq \{x \in \mathcal{X} \ominus \mathcal{Z}(H_p) : K_f(\theta(H_p - 1))x \in \mathcal{U} \ominus K(\theta)\mathcal{Z}(H_p - 1)\} \quad (50)$$

Remark 6. Increasing the prediction horizon significantly raises the risk of reaching empty sets, leading to an optimal problem that cannot be solved.

Assumption 10 ([28]). We consider the following assumptions on the MPC cost function:

1. $\mathcal{L}(x, u)$ is uniformly continuous in $\mathbb{Z}$, with the $\mathcal{K}$ -function $L_c(\cdot)$ i.e. for all $x_1, x_2 \in \mathbb{X}$ and $u \in \mathbb{U}$, such that the following holds

$$\|L(x_1, u) - L(x_2, u)\| \leq L_c(\|x_1 - x_2\|), \quad (51)$$

2. $V_f(x)$ is uniformly continuous in $\mathbb{Z}$, with the $\mathcal{K}$ -function L_v i.e. for all $x_1, x_2 \in \mathbb{Z}$, such that the following holds

$$\|V(x_1) - V(x_2)\| \leq L_v(\|x_1 - x_2\|). \quad (52)$$

At this stage, in order to analyse the asymptotic stability of the Robust EMPC-based closed-loop system, the following rotated cost function with terminal costs is considered:

$$\mathcal{L}(\bar{x}, \bar{u}) = \ell(\bar{x}, \bar{u}) - \ell(\bar{x}_s^*, \bar{u}_s^*) + \gamma(\bar{x}) - \gamma(A(\theta(k))\bar{x}(k) + B(\theta(k))\bar{u}(k)) \quad (53a)$$

$$\bar{V}_f := V_f(\bar{x}) - V_f(\bar{x}_s^*) + \gamma(\bar{x}) - \gamma(\bar{x}_s^*) \quad (53b)$$

with the costs $\bar{V}_f$ and $\mathcal{L}$, the following auxiliary OCP is introduced:

$$\min_{\bar{x}(k) \ominus \bar{u}(k)} \bar{L}(\bar{x}(k), \bar{u}(k)) := \sum_{n=0}^{H_p-1} \mathcal{L}(\bar{x}(i), \bar{u}(i)) + \bar{V}_f(\bar{x}(H_p)) \quad (54)$$

subject to (43b)–(43f)

since constraints in problem (54) are the same as in problem (43), both problems have identical feasible set for all $k \in \mathbb{I}_{\geq 0}$. And consequently, solutions exist for both problems for $x \in \mathcal{X}(H_p)$.

Remark 7. According to Assumption 7, it can be assumed that $\gamma(\bar{x}_s^*) = 0$ for all $k \in \mathbb{I}_{\geq 0}$ without loss of generality.

Lemma 2. Consider Assumptions 5–7 hold. The solution of the auxiliary problem (54) is identical to the solution of the original problem (43).

Proof. By considering that both problems only differ in the cost function, expanding the rotated cost function yields

$$\begin{aligned}
 \bar{L}(\bar{\mathbf{x}}(k), \bar{\mathbf{u}}(k)) &= \sum_{i=0}^{H_p-1} \mathcal{L}(\bar{\mathbf{x}}(i), \bar{\mathbf{u}}(i)) + \bar{V}_f(x_{H_p}), \\
 &= \sum_{i=0}^{H_p-1} \ell(\bar{\mathbf{x}}(i), \bar{\mathbf{u}}(i)) - \ell(\bar{\mathbf{x}}_s^*, \bar{\mathbf{u}}_s^*) + \gamma(\bar{\mathbf{x}}) - \gamma(A(\theta(i))\bar{\mathbf{x}}_i + B(\theta(i))\bar{\mathbf{u}}(i)) \\
 &\quad + V_f(\bar{\mathbf{x}}_{H_p}) - V_f(\bar{\mathbf{x}}_s^*) + \gamma(\bar{\mathbf{x}}_{H_p}) - \gamma(\bar{\mathbf{x}}_s^*), \\
 &= L(\bar{\mathbf{x}}(k), \bar{\mathbf{u}}(k)) - V_f(\bar{\mathbf{x}}_s^*) + \gamma(\bar{\mathbf{x}}_{H_p}) - \gamma(\bar{\mathbf{x}}_s^*) + \gamma(\bar{\mathbf{x}}) - \gamma(\bar{\mathbf{x}}_{H_p}) - \sum_{i=0}^{H_p-1} \ell(\bar{\mathbf{x}}_s^*, \bar{\mathbf{u}}_s^*).
 \end{aligned} \tag{55}$$

Subsequently, according to Remark 4, it is obtained that

$$\bar{L}(\bar{\mathbf{x}}(k), \bar{\mathbf{u}}(k)) = L(\bar{\mathbf{x}}(k), \bar{\mathbf{u}}(k)) + \gamma(\bar{\mathbf{x}}) - \sum_{i=0}^{H_p-1} \ell(\bar{\mathbf{x}}_s^*, \bar{\mathbf{u}}_s^*). \tag{56}$$

Since $\sum_{i=0}^{H_p-1} \ell(\bar{\mathbf{x}}_s^*, \bar{\mathbf{u}}_s^*)$ and $\gamma(\bar{\mathbf{x}})$ are independent of the decision variable $\bar{\mathbf{u}}$ for a given initial state $x \in \mathcal{X}$, functions $\bar{L}(\bar{\mathbf{x}}(k), \bar{\mathbf{u}}(k))$ and $L(\bar{\mathbf{x}}(k), \bar{\mathbf{u}}(k))$ differ only by a constant. Therefore, the optimization problems in (43) and (54) have the same identical solutions for all time steps $k \in \mathbf{I}_+$. This ends the proof. $\square$

Lemma 3. The modified cost $\mathcal{L}$ and $\bar{V}_f$ satisfy the property

$$\bar{V}_f((A(\theta(k)) + B(\theta(k))K_f(\theta(k)))\bar{\mathbf{x}}(k)) \leq \bar{V}_f(\bar{\mathbf{x}}) - \mathcal{L}(\bar{\mathbf{x}}, K_f(\theta(k))\bar{\mathbf{x}}) \tag{57}$$

if and only if $V_f(k)$ and $\ell(k)$ satisfy Assumption 7.

Proof 1. Considering Remarks 4 and 5 and performing the proper manipulation of (53b) by adding the term $\gamma(\bar{\mathbf{x}}) + \gamma((A(\theta(k)) + B(\theta(k))K_f(\theta(k)))\bar{\mathbf{x}}(k))$ to both sides. $\square$

Lemma 4 ([29]). Let $\rho(x) : C \rightarrow \mathbb{R}_{\geq 0}$ be a positive definite function that is defined on the compact set C . Then, there exists a class $\mathcal{K}$ function $\zeta(\cdot)$ such that

$$\rho(x) \geq \zeta(\|x\|), \forall x \in C. \tag{58}$$

Lemma 5. Let Assumptions 4–7 hold. The terminal cost $\bar{V}_f$ and the rotated stage cost $\mathcal{L}$ satisfy the following inequalities:

$$\mathcal{L}(\bar{\mathbf{x}}, \bar{\mathbf{u}}) \geq \zeta(\|\bar{\mathbf{x}} - \bar{\mathbf{x}}_s^*\|) \geq 0, \quad \forall (\bar{\mathbf{x}}, \bar{\mathbf{u}}) \in \bar{\mathcal{X}} \times \bar{\mathcal{U}} \tag{59}$$

$$\zeta(\|\bar{\mathbf{x}} - \bar{\mathbf{x}}_s^*\|) \leq \bar{V}_f \leq \hat{\zeta}(\|\bar{\mathbf{x}} - \bar{\mathbf{x}}_s^*\|), \quad \forall (\bar{\mathbf{x}}) \in \mathcal{X}_f \ominus \mathcal{Z} \tag{60}$$

where functions $\zeta(\cdot)$ and $\hat{\zeta}(\cdot)$ are of class $\mathcal{K}_\infty$.

Proof. From (47), (53a) and Assumption 4 it holds that $\mathcal{L}(\bar{\mathbf{x}}, \bar{\mathbf{u}}) \geq \rho(\|\bar{\mathbf{x}} - \bar{\mathbf{x}}_s^*\|)$ for all $\bar{\mathbf{x}} \in \bar{\mathcal{X}}$ and $\bar{\mathbf{u}} \in \bar{\mathcal{U}}$, which according to Lemma 4, leads to (59). It can be shown from (57) and (59) that $\bar{V}_f(\bar{\mathbf{x}}) \geq \sum_{i=0}^{\infty} \mathcal{L}(\bar{\mathbf{x}}(i), K_f(\theta(k))\bar{\mathbf{x}}(i))$. Moreover, from Assumption 5, $\bar{V}_f(x(k))$ is bounded and by $\bar{V}_f(\bar{\mathbf{x}}_s^*) = 0$, thus, it can be upper bounded by a class $\mathcal{K}_\infty$ function, that is $\bar{V}_f(\bar{\mathbf{x}}) \leq \hat{\zeta}(\|\bar{\mathbf{x}} - \bar{\mathbf{x}}_s^*\|)$ for all $x \in \mathcal{X}_f$. This ends the proof. $\square$

Definition 6 ([18]). A continuous function $V : \mathcal{R}^n \rightarrow \mathbb{R}_+$ is called iss-Lyapunov function for system (6) if there exists a class $\mathcal{K}_\infty$ functions $\alpha_1(\cdot)$, $\alpha_2(\cdot)$ and $\alpha_3(\cdot)$, and a class $\mathcal{K}$ function $\alpha_4(\cdot)$ such that the following holds

$$\alpha_1(\|x\|) \leq V(x(k)) \leq \alpha_2(\|x\|), \tag{61a}$$

$$V(x(k+1)) - V(x(k)) \leq -\alpha_3(\|x\|) + \alpha_4(\|w\|), \tag{61b}$$

Theorem 2 (recursive feasibility). Consider a system described by (6), and let Assumptions 4–10 hold, and let $\bar{\mathbf{x}}(k)$ and $\bar{\mathbf{u}}(k)$ denote a feasible solution to the optimization problem of the robust economic model predictive control based on linear parameter varying (REPMC-LPV) formulation (43) at time step k , then the closed-loop system satisfies all the constraints and the subsequent state $\bar{\mathbf{x}}(k+1)$ and $\bar{\mathbf{u}}(k+1)$ will also be feasible for all possible disturbances $w(k) \in \mathcal{W}$.

Proof. This theorem is an extension of the Theorem 2 in [15].

The optimal solution of the REPMC-LPV (43) at time instant k is given by $\bar{\mathbf{u}}^*(k)$ together with its optimal predicted states sequence $\bar{\mathbf{x}}^*(k)$.

$$\bar{\mathbf{u}}^*(k) = [\bar{u}^*(0|k), \bar{u}^*(1|k) \dots \bar{u}^*(H_p - 1|k)]^T,$$

$$\bar{\mathbf{x}}^*(k) = [\bar{x}^*(0|k), \bar{x}^*(1|k) \dots \bar{x}^*(H_p|k)]^T,$$

Denoting a candidate input sequence and its corresponding state sequence for time step $k + 1$ such that $x(k + 1) = A(\theta(k))x(k) + B(\theta(k))\bar{u}^*(0|k) + w(k)$ as follows:

$$\begin{aligned}\bar{\mathbf{u}}(k + 1) &= [\bar{u}(0|k + 1), \bar{u}(1|k + 1) \dots \bar{u}(H_p - 1|k + 1)]^T, \\ \bar{\mathbf{x}}(k + 1) &= [\bar{x}(0|k + 1), \bar{x}(1|k + 1) \dots \bar{x}(H_p|k + 1)]^T,\end{aligned}$$

Additionally let

$$\epsilon(i) = \bar{x}(i|k + 1) - \bar{x}^*(i + 1|k), \quad \forall i \in [0 \dots H_p - 1] \quad (62)$$

we construct a candidate solution as follows, first

$$\bar{x}(0|k + 1) = x(k + 1),$$

Then, take

$$\bar{u}(i|k + 1) = \bar{u}^*(i + 1|k) + K(\theta(i|k + 1))\epsilon(i), \quad \forall i \in [1 \dots H_p - 2], \quad (63)$$

$$\bar{x}(i|k + 1) = A(\theta(i - 1|k + 1))\bar{x}(i - 1|k + 1) + B(\theta(i - 1|k + 1))\bar{u}(i - 1|k + 1), \quad \forall i \in [1 \dots H_p],$$

where

$$\bar{u}(H_p - 1|k + 1) = K_f(\theta(H_p - 1|k + 1))\bar{x}(H_p - 1|k + 1), \quad i = H_p - 1$$

In the following we show that the candidate solution is a feasible solution at time step $k + 1$ for any $w(k) \in \mathcal{W}$. Indeed from the definition of (62) we have that:

$$\begin{aligned}\epsilon(i + 1) &= \bar{x}(i + 1|k + 1) - \bar{x}^*(i + 2|k), \\ &= A(\theta(i|k + 1))\bar{x}(i|k + 1) + B(\theta(i|k + 1))\bar{u}(i|k + 1) - \\ &\quad (A(\theta(i + 1|k))\bar{x}^*(i + 1|k) + B(\theta(i + 1|k))\bar{u}^*(i + 1|k)), \\ &= A(\theta(i|k + 1))\bar{x}(i|k + 1) + B(\theta(i|k + 1))(\bar{u}^*(i + 1|k) + K(\theta(i|k + 1))\epsilon(i)) - \\ &\quad (A(\theta(i + 1|k))\bar{x}^*(i + 1|k) + B(\theta(i + 1|k))\bar{u}^*(i + 1|k)), \\ &= (A(\theta(i|k + 1)) + B(\theta(i|k + 1))K(\theta(i|k + 1))) \underbrace{(\bar{x}(i|k + 1) - \bar{x}^*(i + 1|k))}_{\epsilon(i)} + \\ &\quad \underbrace{(A(\theta(i|k + 1)) - A(\theta(i + 1|k)))\bar{x}^*(i + 1|k)}_{\delta A(i)} + \underbrace{(B(\theta(i|k + 1)) - B(\theta(i + 1|k)))\bar{u}^*(i + 1|k)}_{\delta B(i)}, \\ &= (A(\theta(i + 1|k)) + B(\theta(i + 1|k))K(\theta(i + 1|k)))\epsilon(i) + \underbrace{\delta A(i)\bar{x}^*(i + 1|k) + \delta B(i)\bar{u}^*(i + 1|k)}_{w_r(i)},\end{aligned} \quad (64)$$

$$\epsilon(i + 1) = A_K(\theta(i + 1|k))\epsilon(i) + w_r(i), \quad \forall i \in [0 \dots H_p - 2],$$

Then, noting that

$$\epsilon(0) = x(k + 1) - (A(\theta(k))x(k) + B(\theta(k))u(k)) = w_d(k), \quad (65)$$

And recursively applying (64), we obtain

$$\bar{x}(i|k + 1) - \bar{x}^*(i + 1|k) = \oplus_{j=0}^i A_K^j(\theta(i))w(k), \quad \forall i \in [0 \dots H_p - 1], \quad (66)$$

By definition of the RPI set we have that

$$\bar{x}(i|k + 1) \in \bar{x}^*(i + 1|k) \oplus \mathcal{Z}(i + 1), \quad \forall i \in [0 \dots H_p - 1], \quad (67)$$

Therefore

$$\bar{x}(i|k + 1) \subset (\mathcal{X} \ominus \mathcal{Z}(i + 1)) \oplus \mathcal{Z}(i + 1), \quad (68)$$

And hence

$$\begin{aligned}\bar{x}(i|k + 1) &\subseteq ((\mathcal{X} \ominus \mathcal{Z}(i)) \ominus \mathcal{Z}(i + 1)) \oplus \mathcal{Z}(i + 1), \\ &\subseteq \mathcal{X} \ominus \mathcal{Z}(i), \quad \forall i \in [0 \dots H_p - 1],\end{aligned} \quad (69)$$

Which proves the satisfaction of (43d).

From (63) and taking into account (62) and (66), we have

$$\bar{u}(i + 1|k + 1) = \bar{u}^*(i + 1|k) + K(\theta(i))\epsilon(i + 1),$$

Which, following the same procedure used before

$$\begin{aligned}\bar{u}(i|k+1) &\in \bar{u}^*(i+1|k) \oplus K(\theta(i+1))\mathcal{Z}(i+1), \\ &\subset (\mathcal{U} \ominus K(\theta(i+1))\mathcal{Z}(i+1)) \oplus K(\theta(i+1))\mathcal{Z}(i+1), \\ &\subseteq ((\mathcal{U} \ominus K(\theta(i))\mathcal{Z}(i)) \ominus K(\theta(i+1))\mathcal{Z}(i+1)) \oplus K(\theta(i+1))\mathcal{Z}(i+1), \\ &\subseteq \mathcal{U} \ominus K(\theta(i))\mathcal{Z}(i), \quad \forall i \in [0 \dots H_p - 2],\end{aligned}\tag{70}$$

Next taking $i = H_p - 1$ in (67) leads to

$$\begin{aligned}\bar{x}(H_p - 1|k+1) &\in \bar{x}^*(H_p|k) \oplus \mathcal{Z}(i), \\ &\subset (\mathcal{X}_f \ominus \mathcal{Z}(i)) \oplus \mathcal{Z}(i), \\ &\subseteq \mathcal{X}_f.\end{aligned}\tag{71}$$

Therefore, taking into account the definition of the pontryagin difference, we have that

$$\begin{aligned}\bar{x}(H_p|k+1) &= A(\theta(H_p - 1|k+1))\bar{x}(H_p - 1|k+1) + B(\theta(H_p - 1|k+1))u(H_p - 1|k+1), \\ &= (A(\theta(H_p - 1|k+1)) + B(\theta(H_p - 1|k+1))K_f(\theta(H_p - 1|k+1)))\bar{x}(H_p - 1|k+1), \\ &\subseteq (A(\theta(H_p - 1|k+1)) + B(\theta(H_p - 1|k+1))K_f(\theta(H_p - 1|k+1)))\mathcal{X}_f, \\ &\subseteq \mathcal{X}_f \ominus \mathcal{Z}(H_p),\end{aligned}\tag{72}$$

Which proofs the satisfaction of (43f).

Finally, since $\bar{x}(H_p - 1|k+1) \in \mathcal{X}_f$, and taking into account Assumption 9, we have that

$$\begin{aligned}\bar{u}(H_p - 1|k+1) &= K_f(\theta(H_p - 1|k+1))\bar{x}(H_p - 1|k+1), \\ &\in \mathcal{U} \ominus K(\theta(H_p - 1))\mathcal{Z}(H_p - 1),\end{aligned}\tag{73}$$

which alongside (70) proofs the satisfaction of (43e). This completes the proof. $\square$

Theorem 3 (Input to state stability). *Let Assumptions 4–10 hold. Then the closed loop system of (6) is ISS for all feasible initial states, with respect to all possible disturbances $w(k) \in \mathcal{W}$. The candidate Lyapunov function $\bar{L}^0(x(k))$ satisfies*

$$\zeta(\|x - x_s^*\|) \leq \bar{L}^0(x(k)) \leq \hat{\zeta}(\|x - x_s^*\|),\tag{74}$$

$$\bar{L}^0(x(k+1)) - \bar{L}^0(x(k)) \leq \gamma(\|w\|) - \underline{\zeta}(\|x - x_s^*\|),\tag{75}$$

for all $x \in \mathcal{X}_f$, $w \in \mathcal{W}$, where $\zeta(\cdot)$, $\underline{\zeta}(\cdot)$ and $\hat{\zeta}(\cdot)$ being class $\mathcal{K}_\infty$ functions, and $\gamma(\cdot)$ is a class $\mathcal{K}$ function.

Proof. This theorem is an extension of Theorem 2 in [15].

The optimal solution of the REMPC (43) at time instant k is given by $\bar{\mathbf{u}}^*(k)$ together with its optimal predicted states sequence $\bar{\mathbf{x}}^*(k)$.

$$\begin{aligned}\bar{\mathbf{u}}^*(k) &= [\bar{u}^*(0|k), \bar{u}^*(1|k), \dots, \bar{u}^*(H_p - 1|k)]^T, \\ \bar{\mathbf{x}}^*(k) &= [\bar{x}^*(0|k), \bar{x}^*(1|k), \dots, \bar{x}^*(H_p|k)]^T,\end{aligned}$$

Denoting a candidate input sequence and its corresponding state sequence for time step $k+1$ such that $x(k+1) = A(\theta(k))x(k) + B(\theta(k))\bar{u}^*(0|k) + w(k)$ as follows:

$$\begin{aligned}\bar{\mathbf{u}}(k+1) &= [\bar{u}(0|k+1), \bar{u}(1|k+1) \dots \bar{u}(H_p - 1|k+1)]^T, \\ \bar{\mathbf{x}}(k+1) &= [\bar{x}(0|k+1), \bar{x}(1|k+1) \dots \bar{x}(H_p|k+1)]^T,\end{aligned}$$

based on the optimal solutions at time step k we have :

$$\bar{\mathbf{u}}(k+j|k) = \begin{cases} u^*(i+1|k), & \text{for } i \in 0, \dots, H_p - 2 \\ K_f(\bar{x}(H_p - 1|k+1)), & \text{for } i = H_p - 1 \end{cases}\tag{76}$$

and hence:

$$\bar{\mathbf{u}}(k+1) = [\bar{u}^*(1|k), \bar{u}^*(2|k) \dots \bar{u}^*(H_p - 1|k), K_f(\bar{x}(H_p - 1|k))]^T,$$

where $\bar{x}(H_p + 1|k) = (A(\theta(k)) + B(\theta(k))K_f(\bar{x}(H_p - 1|k)))\bar{x}(H_p|k+1)$.

The optimal cost of the REMPC-LPV optimization problem (43) at time step k is given by

$$\bar{L}^0(\bar{\mathbf{x}}^*(k), \bar{\mathbf{u}}^*(k)) = \sum_{i=0}^{H_p-1} \mathcal{L}(\bar{x}^*(i|k), \bar{u}^*(i|k)) + \bar{V}_f(\bar{x}^*(H_p|k))$$

And the cost of the feasible sequence at time step $k + 1$ is as follows

$$\bar{L}(\bar{x}(k+1), \bar{u}(k+1)) = \sum_{i=0}^{H_p-1} \mathcal{L}(\bar{x}(i|k+1), \bar{u}(i|k+1)) + \bar{V}_f(\bar{x}(H_p|k+1))$$

Taking the difference between them leads to

$$\Delta \bar{L} = \bar{L}(\bar{x}(k+1), \bar{u}(k+1)) - \bar{L}(\bar{x}(k), \bar{u}(k)),$$

$$\Delta \bar{L} = \sum_{i=0}^{H_p-1} \mathcal{L}(\bar{x}(i|k+1), \bar{u}(i|k+1)) - \sum_{i=0}^{H_p-1} \mathcal{L}(\bar{x}(i|k), \bar{u}(i|k)) + \bar{V}_f(\bar{x}(H_p|k+1)) - \bar{V}_f(\bar{x}(H_p|k)).$$

from the definition of the candidate solution at time step $k + 1$ we have that:

$$\begin{aligned} \Delta \bar{L} &= \bar{V}_f(\bar{x}(H_p|k+1)) - \bar{V}_f(\bar{x}(H_p|k)) + \mathcal{L}(\bar{x}(H_p-1|k+1), K_f(\bar{x}(H_p-1|k+1))) - \mathcal{L}(x(k), \bar{u}^*(0|k)) + \\ &\quad + \sum_{i=0}^{H_p-2} \{ \mathcal{L}(\bar{x}(i|k+1), \bar{u}^*(i+1|k)) - \mathcal{L}(\bar{x}(i+1|k), \bar{u}^*(i+1|k)) \} \\ \Delta \bar{L} &= \bar{V}_f(\bar{x}(H_p|k+1)) - \bar{V}_f(\bar{x}(H_p|k)) + \mathcal{L}(\bar{x}(H_p-1|k+1), K_f(\bar{x}(H_p-1|k+1))) - \mathcal{L}(x(k), \bar{u}^*(0|k)) + \\ &\quad + \sum_{i=0}^{H_p-2} \{ \mathcal{L}(\bar{x}(i|k+1), \bar{u}^*(i+1|k)) - \mathcal{L}(\bar{x}(i+1|k), \bar{u}^*(i+1|k)) \} + \bar{V}_f(\bar{x}(H_p-1|k+1)) - \bar{V}_f(\bar{x}(H_p-1|k+1)) \end{aligned}$$

The C^1 property of the nonlinear model ensures the uniform continuity. Therefore, drawing from similar arguments presented in a related study [28], there exists a class $\mathcal{K}$ -function $\sigma_x(\cdot)$ such that $\|\bar{x}(i|k+1) - \bar{x}(i+1|k)\| \leq \sigma_x^i(\|w(k)\|)$. Hence from the definition of the feasible solution of the control sequence at the next time step in (76) we have that $\bar{u}(i|k+1) = \bar{u}^*(i+1|k)$, then using Theorem 1 and Assumption 10 we get:

$$\mathcal{L}(\bar{x}(i|k+1), \bar{u}^*(i+1|k)) - \mathcal{L}(\bar{x}(i+1|k), \bar{u}^*(i+1|k)) \leq$$

$$\|\mathcal{L}(\bar{x}(i|k+1), \bar{u}^*(i+1|k)) - \mathcal{L}(\bar{x}(i+1|k), \bar{u}^*(i+1|k))\| \leq L_c(\sigma_x^i(\|w(k)\|)), \quad \forall i \in [0, H_p-2]$$

and analogously we have

$$\bar{V}_f(\bar{x}(H_p-1|k+1)) - \bar{V}_f(\bar{x}(H_p|k+1)) \leq L_v(\sigma_x^{H_p-1}(\|w(k)\|)),$$

$$\begin{aligned} \Delta \bar{L} &\leq (\mathcal{L}(\bar{x}(H_p-1|k+1), K_f(\bar{x}(H_p-1|k+1))) + \bar{V}_f(\bar{x}(H_p|k+1)) - \bar{V}_f(\bar{x}(H_p-1|k+1))) \\ &\quad + \theta_i(\|w(k)\|) - \mathcal{L}(x(k), \bar{u}^*(0|k)), \quad \forall i \in [0, H_p-1] \end{aligned}$$

where $\theta_i(\|w(k)\|) = \sum_{i=0}^{H_p-2} L_c(\sigma_x^i(\|w(k)\|)) + L_v(\sigma_x^{H_p-1}(\|w(k)\|))$ is a class $\mathcal{K}$ -function.

We have from Assumption 7 :

$$\bar{V}_f(\bar{x}(H_p|k+1)) - \bar{V}_f(\bar{x}(H_p-1|k+1)) \leq -\mathcal{L}(\bar{x}(H_p-1|k+1), K_f(\bar{x}(H_p-1|k+1))),$$

And hence:

$$\Delta \bar{L} \leq \theta_i(\|w(k)\|) - \mathcal{L}(x(k), \bar{u}^*(0|k)), \quad \forall i \in [0, H_p-1] \quad (77)$$

From Lemma 5 and 77, it follows that:

$$\bar{L}^0(x(k+1)) - \bar{L}^0(x(k)) \leq \Delta \bar{L} \leq \theta_i(\|w(k)\|) - \underline{\zeta}(\|x - x_s^*\|), \quad (78)$$

which completes the proof. $\square$

3.1. Computation of the local controller

In this section, our aim is to design a state feedback LPV controller to minimize the disparity between the system states (6) and the nominal state vectors (13), even under the presence of exogenous disturbances and model uncertainty.

H_∞ control has been widely used and considered as an efficient control strategy when dealing with uncertainties. Hence, the local controller aims to reduce the deviation between the reference and the real states of the vehicle. This mismatch depends on the model uncertainty and external disturbances that affect the model.

As reported in [30] the linear quadratic regulator (LQR) control strategy is commonly used when determining a local control structure to robustify the MPC strategy using the tube-based approach. however, bu using H_∞ control, there is even greater flexibility to include additional performance weights and better attenuate unknown inputs (disturbance and noise). When dealing with systems subject to external disturbances, the LQR technique becomes less efficient against such system variations than the H_∞ strategy. Thus, it seems more compelling for the considered application.

In this study, the controller gain $K(\theta(k))$ is a gain scheduling H_∞ -LPV state feedback controller with the objective of minimizing the infinity norm between the exogenous disturbances and the performance output.

Given

$$\begin{aligned} x(k+1) &= A(\theta(k))x(k) + B(\theta(k))u(k) + Ed(k) \\ z(k) &= Cx(k) + D_1u(k) + D_2d(k) \end{aligned} \quad (79)$$

where $A(\theta(k))$ and $B(\theta(k))$ are the system dynamics and the input matrix respectively, the matrix E is employed to model the disturbance input, while the vector d signifies the exogenous disturbance. The product of $Ed(k)$ consistently resides within the set $\mathcal{W}$. Meanwhile, the vector z represents the variables under control, and matrices C , D_1 , and D_2 are appropriately sized tuning matrices.

Hence, the proposed problem consists on finding an LPV state feedback gain $K(\theta(k))$ that minimises the H_∞ norm of the transfer between z and d such that:

$$\|G_{zd}\|_\infty \leq \gamma \quad (80)$$

This problem is solved using the following LMIs [31]:

$$\min_{X, W_i} \gamma, \quad (81a)$$

subject to

$$X > 0, \quad (81b)$$

$$\begin{bmatrix} \tilde{X} & A(\theta(k))\tilde{X} - B(\theta(k))W & E & 0 \\ * & \tilde{X} & 0 & \tilde{X}C^T - W^T D_2^T \\ * & * & \gamma I & D_2^T \\ * & * & * & \gamma I \end{bmatrix} > 0, \quad (81c)$$

being the solutions $\tilde{X} = \tilde{P}^{-1} \in \mathcal{R}^{n \times n}$ and $W(\theta(k)) = K(\theta(k))\tilde{P}^{-1} \in \mathcal{R}^{m \times n}$, where $\tilde{P} \in \mathcal{R}^{n \times n} > 0$ represents the common Lyapunov matrix for the LPV system, and the H_∞ norm of G_{zd} is γ . $A(\theta(k))$ is given by :

$$A(\theta(k)) = \sum_{i=1}^N \mu_i(\theta) A_i, \quad (82)$$

where A_i captures the system dynamics at each vertex of the polytope Θ . The total number of vertices in the polytope Θ , denoted by N , is 2^{n_θ} . The linear membership function $\mu_i(\theta)$ is given by:

$$\mu_i(\theta) = \prod_{j=1}^{n_\theta} \xi_{ij}(\eta_0^j, \eta_1^j), \quad \forall i = 1, \dots, N, \quad (83)$$

where:

$$\begin{aligned} \eta_0^j &= \frac{\bar{\theta}_j - \theta_j(k)}{\bar{\theta}_j - \underline{\theta}_j}, \\ \eta_1^j &= 1 - \eta_0^j. \end{aligned} \quad (84)$$

Each scheduling variable θ_j must satisfy condition (7). The function ξ_{ij} covers all $N = 2^{n_\theta}$ possible combinations. Similarly, the control gain $K(\theta(k))$ is also a polytopic controller and is given by:

$$K(\theta(k)) = \sum_{i=1}^N \mu_i(\theta) K_i, \quad (85)$$

Noting that, the control gain $K(\theta(k))$ must satisfy [Assumption 8](#).

3.2. Computation of the terminal components

Assumption 11. There exist a state-feedback controller $K_i(\theta(k))$, such that the closed-loop system is stable for all $\theta \in \Theta$.

$$\tilde{A}(\theta(k)) = A(\theta(k)) + B(\theta(k))K_i(\theta(k)) \quad (86)$$

The computation of the terminal components of (43) requires in addition a terminal control law, which is determined as follows:

$$K_f(\theta(k)) = K_i(\theta(k))(\bar{x}(k) - \bar{x}_s^*) + \bar{u}_s^* \quad (87)$$

where $K_i(\theta(k))$ is the feedback gain satisfying [Assumption 11](#), and the pair $(\bar{x}_s^*, \bar{u}_s^*)$ are elements of the optimal steady state and input trajectories. The terminal penalty is introduced based on the ellipsoidal level sets correlated to quadratic functions of the form

$$V_f(\bar{x}) := \frac{1}{2}(\bar{x} - \bar{x}_s^*)^T P(\bar{x} - \bar{x}_s^*). \quad (88)$$

Therefore, the terminal regions are centered around the optimal state pair $(\bar{x}_s^*, \bar{u}_s^*)$ and defined as

$$\mathcal{X}_f \ominus \mathcal{Z}(H_p) := \{\bar{x} \in \mathbb{R}^n | (\bar{x} - \bar{x}_s^*)^\top P(\bar{x} - \bar{x}_s^*) \leq \beta\} \quad (89)$$

where $\beta \in \mathbb{R}_+$. The scalar β must ensure that the state and input constraints are perpetually satisfied under the use of terminal controller (87), that is, $\bar{x}(k) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p) \subset \bar{\mathcal{X}}$ and $K_f(\theta(k))\bar{x}(k) \in \bar{\mathcal{U}}$ for all $k \in \mathbb{I}_+$. To derive an appropriate terminal function $V_f(\bar{x})$ for the economic cost based on the LPV model in (13), the procedure in reference [32] is suitably modified. First, assume that the economic costs $\ell(\cdot)$ are twice continuously differentiable and let $\bar{\ell}(\bar{x}(k)) := \ell(x(k), K_f(\theta(k))\bar{x}(k)) - \ell(\bar{x}_s^*, \bar{u}_s^*)$. Then from reference [32], for all $x \in \mathcal{X}$, $(\bar{x}_s^*, \bar{u}_s^*) \in \mathbb{Z}$ and $k \in \mathbb{I}_{[0, H_p-1]}$, there exists a matrix Q such that $Q - \bar{\ell}(\bar{x}(k)) \geq 0$

Moreover, the quadratic cost functional is defined as

$$\ell_q(\bar{x}) := \frac{1}{2}(\bar{x} - \bar{x}_s^*)^\top Q(\bar{x} - \bar{x}_s^*) + q^\top(\bar{x} - \bar{x}_s^*), \quad (90)$$

where $q := \bar{\ell}_x(\bar{x}_s^*(k), \bar{u}_s^*(k))$ such that for all $\bar{x} \in \bar{\mathcal{X}}$ the inequality $\ell_q(\bar{x}) \geq + (1/2)(\bar{x} - \bar{x}_s^*)^\top (\bar{x} - \bar{x}_s^*)$ holds. Hence, the candidate terminal function is defined as

$$V_f := \sum_{i=0}^{\infty} \ell_q(\bar{x}(i)), \quad (91)$$

where $\bar{x}(i+1) = A(\theta(i))\bar{x}(i) + B(\theta(i))(K_f(\theta(i))(\bar{x} - \bar{x}_s^*) + \bar{u}_s^*)$ for all $i \in \mathbb{I}_+$.

Let us obtain an explicit definition of (91). From Assumption 11 and (87), the error dynamics are given by

$$(\bar{x}(k+1) - \bar{x}_s^*) = \tilde{A}(\theta(k))(\bar{x}(k) - \bar{x}_s^*), \quad \forall k \in \mathbb{I}_+. \quad (92)$$

The so-called monodromy matrix of system (92) at time step k is defined as

$$\Psi_k := \prod_{l=0}^k \tilde{A}(\theta(l)), \quad \forall k \in \mathbb{I}_+. \quad (93)$$

Then, according to (90), (92), and (93), the terminal function (91) can be written as follows:

$$\begin{aligned} V_f &= \frac{1}{2} \sum_{l=0}^{n_f-1} (\bar{x}(l) - \bar{x}_s^*)^\top \left(\sum_{i=0}^{\infty} (\Psi_l^i)^\top Q_l (\Psi_l^i) \right) (\bar{x}(l) - \bar{x}_s^*) + \bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top \sum_{i=0}^{\infty} (\Psi_l^i) (\bar{x}(l) - \bar{x}_s^*) \\ &= \frac{1}{2} \sum_{l=0}^{H_p-1} (\bar{x}(l) - \bar{x}_s^*)^\top P_l (\bar{x}(l) - \bar{x}_s^*) + q_l^\top (\bar{x}(l) - \bar{x}_s^*), \end{aligned} \quad (94)$$

with $\bar{x}(l+1) = A(\theta(l))\bar{x}(l) + B(\theta(l))(K_f(\theta(l))(\bar{x}(l) - \bar{x}_s^*) + \bar{u}_s^*)$ for all $l \in \mathbb{I}_{[0, H_p-1]}$ and $q_l^\top = \bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top (I - \Psi_l)^{-1}$. For the given k and each $l \in \mathbb{I}_{[0, H_p-1]}$, matrices P_l are the solutions to the discrete Lyapunov equations $\Psi_l^\top P_l \Psi_l - P_l = -Q_l$. Through (91), it follows that the candidate function V_f satisfies the condition in Assumption 7. In fact, from Assumption 11 and suitable manipulation of (91)–(94), the following balance may be derived:

$$\begin{aligned} V_f((A(\theta(k)) + B(\theta(k))K_f(\theta(k)))\bar{x}(k)) - V_f(x(k)) &= \\ \frac{1}{2}(\bar{x}(k) - \bar{x}_s^*(k))^\top (\tilde{A}(\theta(k))^\top P_k \tilde{A}(\theta(k)) - P_k)(\bar{x}(k) - \bar{x}_s^*(k)) - \bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top (\bar{x}(k) - \bar{x}_s^*). \end{aligned} \quad (95)$$

Hence, (91) and (95) yield the following Lyapunov equation:

$$\Psi_k^\top P_k \Psi_k - P_k = -Q_k, \quad \forall k \in \mathbb{I}_{[0, H_p-1]}. \quad (96)$$

It is important to highlight that matrices P_k and $K_f(\theta(k))$ must satisfy in addition the following condition:

$$\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - P_k \leq 0, \quad \forall k \in \mathbb{I}_{[0, H_p-1]}. \quad (97)$$

to satisfy the positive invariant condition, that is, if $\bar{x}(k) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$ then $(\bar{x}(k), K_f(\theta(k))\bar{x}(k)) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$ for all $k \in \mathbb{I}_{\geq 0}$.

The main problem to do so is that the matrices in (93) lead to nonlinear matrix inequalities that cannot be solved directly. Despite the mentioned difficulty for solving (96), it is still conceivable to obtain a set of matrices satisfying Assumption 11. To do so, condition (96) should be relaxed and considering instead $\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - P_k \leq -Q_k$ for all $k \in \mathbb{I}_{[0, H_p-1]}$. The previous discussion is formalized by means of the result in Theorem 4.

Theorem 4. Consider system (6) controlled by (16), Assumptions 5–9 are satisfied, with the terminal control law (87) and the pair $x_s^*, u_s^* \in \mathbb{Z}$. Let $X_k \in \mathbb{S}_{++}^{n_x}$, $Y_k \in \mathbb{R}^{n_x \times n_u}$, $\beta \in \mathbb{R}_+$ and $v \in \mathbb{R}_+$ be decision variables. By solving

$$\max_{X_k > 0, Y_k \in \mathbb{R}^{n_x \times n_u}, \beta \in \mathbb{R}_+, v \in \mathbb{R}_+} -\log \det(X_0) \quad (98a)$$

subject to

$$\begin{bmatrix} X_k & \star & \star \\ A(\theta(k))X_k + B(\theta(k))Y_k & X_{k+1} & \star \\ Q_k^{1/2}X_k & 0 & \beta I_n \end{bmatrix} > 0, \quad (98b)$$

$$\begin{bmatrix} vX_k & \star & \star \\ 0 & 1-v & \star \\ A(\theta(k))X_k + B(\theta(k))Y_k & d & X_{k+1} \end{bmatrix} > 0, \quad (98c)$$

$$\begin{bmatrix} (\tilde{h}_j - (a_j x_s^* + b_j u_s^*))^2 & \star \\ X_k a_j^\top + Y_k^\top b_j^\top & X_k \end{bmatrix} > 0, \quad (98d)$$

for all $k \in \mathbb{I}_{[0, H_p-1]}$, $i \in \mathbb{I}_{[1, n_r]}$ and $d \in \text{vert}(\mathcal{Z}(H_p))$, the optimal decision variables (denoted by $X_{opt,k}$, $Y_{opt,k}$, β_{opt} and v_{opt}) allow to determine the controller

$$K_f(\theta(k)) := Y_k X_k^{-1}, \quad (99)$$

$$P_k := X_{opt,k}^{-1} \beta_{opt}. \quad (100)$$

$\forall k \in \mathbb{I}_{[0, H_p-1]}$, P_{H_p} allows determining the terminal cost (88) and region (89).

Proof. First, it should be indicated that the solution of (98) presents the terminal sets defined in (89) for system $\bar{x}(k+1) = (A(\theta(k)) + B(\theta(k))K_f(\theta(k)))\bar{x}(k)$ with K_f as defined in (87). To do so, remark that for $k \in \mathbb{I}_+$ the term $\bar{x}(k) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$ is equal to the following quadratic functional condition:

$$F_0 = (\bar{x}(k) - \bar{x}_s^*)^\top P_k (\bar{x}(k) - \bar{x}_s^*) - \beta \leq 0.$$

Similarly, the requirement that $\bar{x}(k+1) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$ is equivalent to

$$F_1 = (\bar{x}(k+1) - \bar{x}_s^*)^\top P_{k+1} (\bar{x}(k+1) - \bar{x}_s^*) - \beta \leq 0.$$

Using the S-procedure according to Reference [33], the term that $\bar{x}(k) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$ implies $\bar{x}(k+1) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$, is equal to the existence of $\zeta > 0$, such that

$$(\bar{x}(k+1) - \bar{x}_s^*)^\top P_{k+1} (\bar{x}(k+1) - \bar{x}_s^*) - \beta - \zeta ((\bar{x}(k) - \bar{x}_s^*)^\top P_k (\bar{x}(k) - \bar{x}_s^*) - \beta) < 0,$$

$$(\bar{x}(k) - \bar{x}_s^*)^\top \tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) (\bar{x}(k) - \bar{x}_s^*) - \beta - \zeta ((\bar{x}(k) - \bar{x}_s^*)^\top P_k (\bar{x}(k) - \bar{x}_s^*) - \beta) < 0,$$

$$(\bar{x}(k) - \bar{x}_s^*)^\top (\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - \zeta P_k) (\bar{x}(k) - \bar{x}_s^*) - \beta + \beta \zeta < 0. \quad (101)$$

As reported in reference [34] the condition (101) can be formulated as a linear matrix inequality as follows

$$\begin{bmatrix} \tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - \zeta P_k & 0 \\ 0 & -\beta + \beta \zeta \end{bmatrix} < 0. \quad (102)$$

notice that the above inequality can be decomposed into two inequalities:

$$0 < \beta \zeta < \beta, \quad (103)$$

$$\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - \zeta P_k < 0. \quad (104)$$

The following inequality $\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - \zeta P_k \leq 0$ is equivalent to the following inequality if and only if there exists a $\zeta > 0$ such that the following holds

$$\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - P_k + Q_k < 0 \quad (105)$$

Applying the schur complement, then Pre- and Post-multiply both sides by $\text{diag}\{P_k^{-1}, I\}$ and using the change of variable (100) we get:

$$\begin{bmatrix} X_k \beta^{-1} - \beta^{-1} X_k Q_k X_k \beta^{-1} & \beta^{-1} X_k \tilde{A}(\theta(k))^\top \\ \tilde{A}(\theta(k)) X_k \beta^{-1} & \beta^{-1} X_{k+1} \end{bmatrix} > 0.$$

Then multiply the above inequality by β and using (99), then applying the schur complement leads to :

$$\begin{bmatrix} X_k & \star & \star \\ A_k(\theta(k))X_k + B_k(\theta(k))Y_k & X_{k+1} & \star \\ Q_k^{1/2} X_k & 0 & \beta I_n \end{bmatrix} > 0 \quad (106)$$

which proofs (98b).

Following the same procedure the LMI (98b) can be obtained by posing the following condition:

$$(\bar{x}(k) - \bar{x}_s^*)^\top P_{k+1} (\bar{x}(k) - \bar{x}_s^*) \leq 1 \implies (\tilde{A}(\theta(k))(\bar{x}(k) - \bar{x}_s^*) + d)^\top P_{k+1} (\tilde{A}(\theta(k))(\bar{x}(k) - \bar{x}_s^*) + d) \leq 1,$$

Which is satisfied by using the S-procedure i.e. if there exists a scalar $v \geq 0$ such that the following holds:

$$(\tilde{A}(\theta(k))(\bar{x}(k) - \bar{x}_s^*) + d)^\top P_{k+1}(\tilde{A}(\theta(k))(\bar{x}(k) - \bar{x}_s^*) + d) + v(1 - (\bar{x}(k) - \bar{x}_s^*)^\top P_{k+1}(\bar{x}(k) - \bar{x}_s^*)) < 1$$

The above inequality can be presented as

$$\begin{bmatrix} (\bar{x}(k) - \bar{x}_s^*)^\top \\ 1 \end{bmatrix} \begin{bmatrix} vP_k - \tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) & -\tilde{A}(\theta(k))^\top P_{k+1} d \\ -d^\top P_{k+1} \tilde{A}(\theta(k)) & \beta(1 - v) - d^\top P_{k+1} d \end{bmatrix} \begin{bmatrix} (\bar{x}(k) - \bar{x}_s^*) \\ 1 \end{bmatrix} < 0,$$

Then applying the Schur complement lemma we get :

$$\begin{bmatrix} vP_k & \star & \star \\ 0 & 1 - v & \star \\ \tilde{A}(\theta(k)) & d & P_{k+1}^{-1} \end{bmatrix} > 0,$$

Then Pre- and Post-multiply both sides by $\text{diag}\{P_k^{-1}, 1, I\}$ and using (99) and (100) we finally get:

$$\begin{bmatrix} vX_k & \star & \star \\ 0 & 1 - v & \star \\ A(\theta(k))X_k + B(\theta(k))Y_k & d & X_{k+1} \end{bmatrix} > 0, \quad (107)$$

Hence condition (49) can be satisfied by finding a feasible solution of (98c).

The terminal set satisfies the state and input constraints for all $k \in \mathbb{I}_{\geq 0}$. This can be shown by retrieving X_k and Y_k according to (98) and applying the Schur complement to (98d), which gives

$$(a_j + b_j K_f(\theta(k)))(\beta P_k^{-1})(a_j + b_j K_f(\theta(k)))^\top \leq (\tilde{h}_j - a_j \bar{x}_s^* - b_j \bar{u}_s^*)^2, \quad (108)$$

with $(\bar{x}, \bar{u}) := \{a_j x + b_j u \leq \tilde{h}_j, \forall j \in \mathbb{I}_{[1, n_p]}\}$. where $\tilde{h}_j$ represent the bounds of the tightened constraints which is computed using zonotopic reachable sets as presented in Section 2.2.

According to [35], this inequality meets the essential condition for an ellipsoid to be enclosed within a polyhedron. Consequently, Eq. (108) ensures that if $\bar{x}(k) \in \mathcal{X}_f \ominus \mathcal{Z}(H_p)$, then the pair $(\bar{x}(k), K_f(\theta(k))\bar{x}(k))$ will also belong to $\mathcal{X}_f \ominus \mathcal{Z}(H_p)$ for all $k \in \mathbb{I}_{\geq 0}$.

To prove (48), which ensures Assumption 7 is met, we pre- and post-multiply (105) by $(\bar{x}(k) - \bar{x}_s^*)^\top$ and $(\bar{x}(k) - \bar{x}_s^*)$, respectively, resulting in the following inequality:

$$(\bar{x}(k) - \bar{x}_s^*)^\top (\tilde{A}(\theta(k))^\top P_{k+1} \tilde{A}(\theta(k)) - P_k)(\bar{x}(k) - \bar{x}_s^*) \leq -(\bar{x}(k) - \bar{x}_s^*)^\top Q_k(\bar{x}(k) - \bar{x}_s^*), \quad (109)$$

for all $k \in \mathbb{I}_+$. By using (92), (93) and summing up (109) from $k = 0$ to $k = H_p - 1$, (109) can be rewritten as

$$(\bar{x}(k) - \bar{x}_s^*)^\top (\Psi_k^\top P_{k+1} \Psi_k - P_k)(\bar{x}(k) - \bar{x}_s^*) \leq - \sum_{k=0}^{H_p-1} (\bar{x}(k) - \bar{x}_s^*(k))^\top Q_k(\bar{x}(k) - \bar{x}_s^*). \quad (110)$$

Then, multiplying (110) by 0.5 and adding $-\bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top (\bar{x}(k) - \bar{x}_s^*)$ to both sides lead to

$$\begin{aligned} & \frac{1}{2} (\bar{x}(k) - \bar{x}_s^*)^\top (\Psi_k^\top P_{k+1} \Psi_k - P_k)(\bar{x}(k) - \bar{x}_s^*) - \bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top (\bar{x}(k) - \bar{x}_s^*) \\ & \leq -\frac{1}{2} \sum_{k=0}^{H_p-1} (\bar{x}(k) - \bar{x}_s^*)^\top Q_k(\bar{x}(k) - \bar{x}_s^*) - \bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top (\bar{x}(k) - \bar{x}_s^*) \\ & \leq -\frac{1}{2} (\bar{x}(k) - \bar{x}_s^*)^\top Q_k(\bar{x}(k) - \bar{x}_s^*) - \bar{\ell}_x(\bar{x}_s^*, \bar{u}_s^*)^\top (\bar{x}(k) - \bar{x}_s^*) \\ & \leq \bar{\ell}(x(k), u(k)) = \ell(\bar{x}(k), K_f \bar{x}(k)) - \ell(\bar{x}_s^*, \bar{u}_s^*), \end{aligned} \quad (111)$$

for all $k \in \mathbb{I}_{\geq 0}$. The second inequality comes from the positive definiteness of Q_k , and the last inequality is obtained from the description of $\ell_q(\bar{x})$ in (90). Hence, according to (94) and (111), condition (48) is satisfied. Consequently, Theorem 4 is proved and the solution of (98) allows to obtain terminal components satisfying Assumption 7. This ends the proof. $\square$

It should be noted that the controller described in (99) is computed online at each iteration of the MPC controller by solving the LMI problem presented in Theorem 4. This controller is completely theoretical and only utilized for computing the terminal conditions necessary to ensure the stability of the REMPC.

3.3. Summary of the REMPC-LPV control strategy

The proposed control strategy consists of two main phases: an off-line design stage and an on-line implementation stage. These phases are summarised in Algorithm 1.

4. Application examples

In this section, we showcase the effectiveness of our REMPC-LPV through two compelling case studies:

Algorithm 1 Implementation procedure of the proposed REMPC-LPV approach.

Off-line Phase:

- 1: Derive the LPV model (6) from the nonlinear system (1) using the nonlinear embedding approach [21].
- 2: Construct the polytopic LPV representation using the bounding box approach (82)–(84).
- 3: Design the local H_∞ -LPV controller by solving the LMI optimisation problem (81).

On-line Phase (executed at each sampling instant k):

- 4: Measure the current system state $x(k)$.
 - 5: Compute the local control gain $K(\theta(k))$ using (85).
 - 6: Evaluate the reachable sets according to (22).
 - 7: Update the state and input constraint sets based on (23).
 - 8: Solve the REMPC-LPV optimisation problem (43).
 - 9: Apply the control law: $u(k) = \bar{u}(k) + K(\theta(k))(x(k) - \bar{x}(k))$.
-

Table 2

Reaction Parameters for Isothermal CSTR Model.

Parameter	Value
c_{Af}	1mol/L
c_{Bf}	0mol/L
V_R	10L
k_r	1.2L/(mol.min)

4.1. Exemple 1: Isothermal CSTR

In this isothermal Continuous Stirred Tank Reactor (CSTR) model, we consider a single first-order, irreversible chemical reaction:



where k is the rate constant. The material balances for the reactor are:

$$\begin{aligned} \frac{dc_A}{dt} &= \frac{Q}{V_R}(c_{Af} - c_A) - k_r c_A + w_1 \\ \frac{dc_B}{dt} &= \frac{Q}{V_R}(c_{Bf} - c_B) + k_r c_A + w_2 \end{aligned} \quad (113)$$

where c_A and c_B are the molar concentrations of A and B respectively, c_{Af} and c_{Bf} are the feed concentrations of A and B , and Q is the flow rate through the reactor. The volume of the reactor is V_R and the rate constant is k_r . An additive disturbance w is assumed to be acting on the system. The parameters for this model are illustrated in Table 2.

The economic objective of this problem is based on the price of the product B and a separation cost, which is assumed to be directly proportional to the flow rate. hence the cost function is given by [20]:

$$\mathcal{L}(c_A, c_B, Q) = -(2Qc_B - (0.5)Q) \quad (114)$$

To ensure strong duality (where the primal and dual problems yield the same optimal value), we adjust the economic objective by adding a quadratic penalty.

$$\mathcal{L}(c_A, c_B, Q) = -(2Qc_A - 0.5Q) + |c_A - \overline{c_{As}}^*|_{Q_A}^2 + |c_B - \overline{c_{Bs}}^*|_Q^2 + |Q - \overline{Q_s}^*|_R^2 \quad (115)$$

The nominal LPV model can be derived using the nonlinear embedding approach as discussed earlier as follows

$$x_{k+1} = A(\theta_k)x_k + B(\theta_k)u_k \quad (116)$$

such that the state and control vectors are as follows:

$$x = \begin{bmatrix} c_A \\ c_B \end{bmatrix}, u = Q. \quad (117)$$

where the scheduling variable θ_k is chosen to be as:

$$\theta_k := Q_k \quad (118)$$

being:

$$A(\theta_k) = \begin{bmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{bmatrix} \quad (119)$$

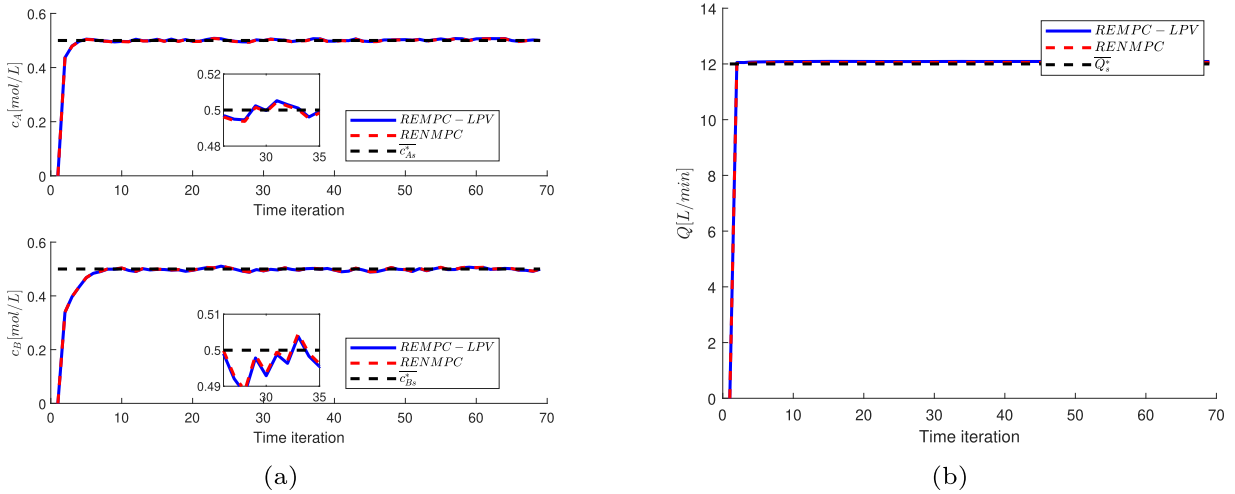


Fig. 1. (a) Tracking the optimal economic reachable steady state for the CSTR. (b) Tracking the optimal economic reachable input for the CSTR.

$$B(\theta_k) = \begin{bmatrix} B_{11} \\ B_{12} \end{bmatrix} \quad (120)$$

where:

$$\begin{aligned} A_{11} &= 1 + T_s(-k_r - \frac{Q}{V_R}), A_{12} = T_s k_r, A_{13} = 1 + T_s(-\frac{Q}{V_R}), \\ B_{11} &= \frac{c_{Af}}{V_R}, B_{12} = \frac{c_{Bf}}{V_R}, \end{aligned} \quad (121)$$

For this simulation, the molar concentrations are constrained within the bounds $0 \leq c_A \leq 1 \text{ mol/L}$ and $0 \leq c_B \leq 1 \text{ mol/L}$. An upper bound constraint is imposed on the flow rate as follows $Q \leq 20 \text{ L/min}$. This model is nonlinear due to the presence of the terms Qc_A and Qc_B . Using (39) the nominal optimal reachable steady state for this cost $\bar{c}_{As}^* = 0.5 \text{ mol/L}$, $\bar{c}_{Bs}^* = 0.5 \text{ mol/L}$ while the optimal reachable input is $\bar{Q}_s^* = 12$. The prediction horizon is set to be $H_p = 30$, and sampling time is for $T_s = 0.5 \text{ min}$.

By solving the H_∞ control problem, we obtain the following positive definite matrix $\tilde{P}$:

$$\tilde{P} = \begin{bmatrix} 11.7870 & 3.1346 \\ 3.1346 & 3.7024 \end{bmatrix} \quad (122)$$

with $\gamma_\infty = 2.2500 \times 10^{-6}$, and the controller gains at the vertices of the scheduling variable are $K_1 = [-0.2676, -0.4303]$ and $K_2 = [-2.7168, -1.8059]$, which then be interpolated using the polytopic approach as (85).

After solving (98) at each iteration, it is obtained that the value of β_{opt} is 0.8134, and v_{opt} is 0.2988. The controller $K_t(\theta(k))$ is determined using $Y_k X_k^{-1}$, and the terminal cost and region are computed based on the terminal value $P_k = X_{opt,k}^{-1} \beta_{opt}$. Hence the optimal terminal controller can be computed by $K_f(\theta(k)) = K_t(\theta(k))(\bar{x}(k) - \bar{x}_s^*) + \bar{u}_s^*$.

Fig. 1a depicts tracking the optimal economic reachable steady state for the CSTR, while Fig. 1b shows tracking the optimal economic reachable input. Both figures demonstrate that solving the REMPC-LPV yields results identical to solving the RENMPC. The tracking error is depicted in Fig. 2a. Fig. 2b shows the disturbances applied to the CSTR system. The zonotopic reachable sets for the CSTR are presented in Fig. 3b. Fig. 3a shows that the closed loop trajectory converges towards the optimal economic concentration.

Table 3 presents a comparative evaluation between the proposed REMPC-LPV and the RENMPC in terms of tracking accuracy and computational efficiency for the CSTR system. The results show that both controllers achieve nearly identical tracking performance, while the REMPC-LPV exhibits a significantly lower computational time per iteration, highlighting its suitability for real-time implementation.

Table 4 presents a quantitative comparison between the RTMPC-LPV and the REMPC-LPV. Both controllers demonstrate comparable computational efficiency, indicating that the incorporation of economic objectives does not introduce a significant computational overhead. In terms of performance, however, a clear distinction is observed in the total cost. This highlights the effectiveness of REMPC-LPV in optimizing economic performance without compromising real-time feasibility (see Fig. 4).

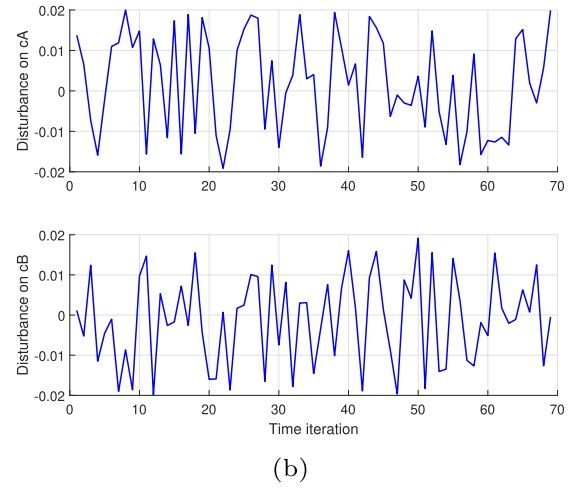
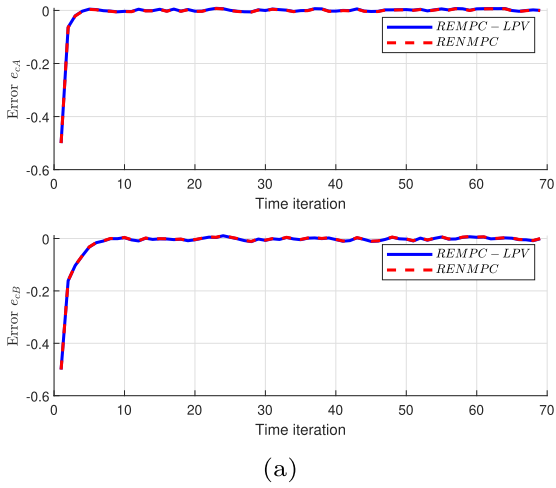
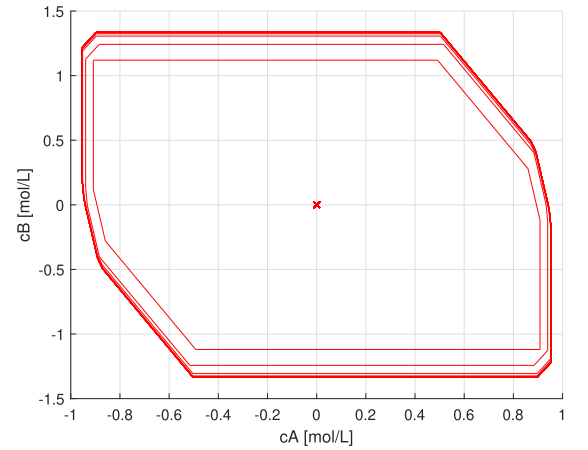
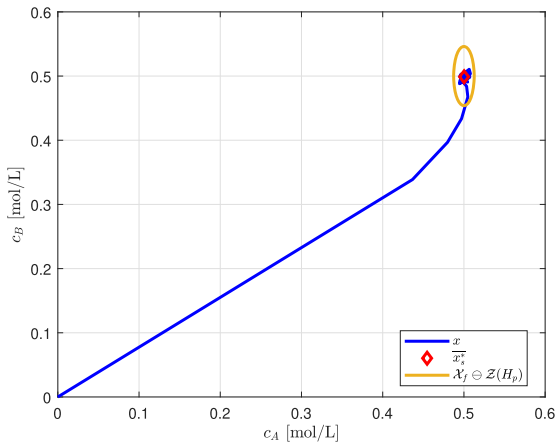


Fig. 2. (a) Tracking error evolution for the CSTR. (b) The disturbance window used for the simulation of the CSTR.



(a) Phase plot of the closed-loop trajectories for the CSTR system.

(b) The zonotopic reachable sets for the CSTR.

Fig. 3. Side by side figures: phase plot and reachable sets for the CSTR.

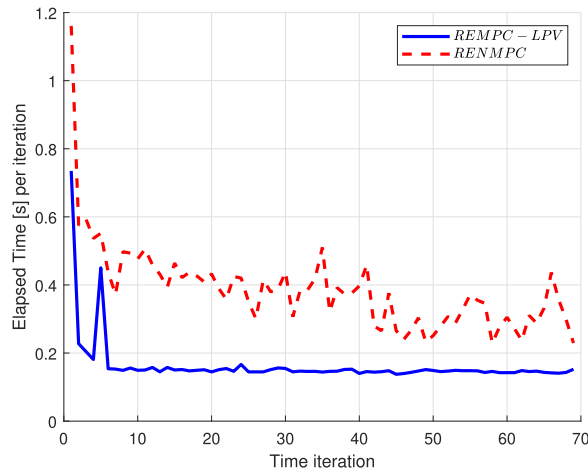


Fig. 4. The elapsed time for solving the control problem for the CSTR.

Table 3

Comparison of error metrics and computational performance for CSTR.

Metric	REMP-CPV	RENMP-CP
Error metrics for c_A		
MSE	0.0037	0.0037
IAE	0.8049	0.8041
ISE	0.2555	0.2555
Error metrics for c_B		
MSE	0.0043	0.0043
IAE	1.1463	1.1323
ISE	0.2941	0.2937
Mean Elapsed Time per Iteration		
Time [s]	0.1632	0.3853

Table 4

Comparison of tracking and economic LPV-MPC performance for the CSTR.

Controller Type	Mean Elapsed Time [s]	Total Cost
RTMPC-LPV	0.1599	404.7428
REMP-CPV	0.1632	-1.9664×10^4

4.2. Exemple 2: Autonomous racing vehicle

4.2.1. Nonlinear vehicle dynamic model

The following nonlinear vehicel dynamic model taken from [38] is considered

$$\begin{aligned}
 \dot{v}_x &= a_r + \frac{-F_{yf} \sin \delta - F_{df}}{m} + \omega v_y - g \sin(\phi) \\
 \dot{v}_y &= \frac{F_{yf} \cos \delta + F_{yr} - F_w}{m} - \omega v_x \\
 \dot{\omega} &= \frac{F_{yf} l_f \cos \delta - F_{yr} l_r - F_w(l_f - l_r)}{I} \\
 F_{yf} &= C_f(\alpha_f)\alpha_f \\
 F_{yr} &= C_r(\alpha_r)\alpha_r \\
 F_{df} &= \mu mg + \frac{1}{2}\rho C_{dA}v_x^2 \\
 F_w &= \frac{1}{2}\rho C_{dAl}v_w^2.
 \end{aligned} \tag{123}$$

where v_x , v_y , and ω represent the body frame velocities linear in x , linear in y , in (m/s), and the angular velocity around the orthogonal axis z to the road, in (rad/s). In turn, the control inputs, δ and a represent the steering angle at the front wheels, in (rad), and the longitudinal acceleration on the rear wheels, in (m/s²), respectively. F_{yf} and F_{yr} represent the lateral forces produced in front and rear tires, being C_f and C_r the tire stiffness coefficient for the front and rear wheels, respectively. Furthermore, m and I represent the vehicle mass and inertia, l_f and l_r represent the distances from the vehicle center of mass toward the front and rear wheel axes, and μ_s and g represent the static friction coefficient and the acceleration due to gravity, respectively. The parameters of this model are given in Table 5. Moreover, the exogenous disturbances are represented by ϕ and v_w , signifying the longitudinal road slope and the lateral wind velocity, respectively.

4.2.2. Nonlinear vehicle curvature based kinematic model

On the other hand, the kinematic model describes the evolution of the vehicle position based on the lateral and orientation errors in the curvilinear reference frame.

$$\begin{aligned}
 \dot{y}_e &= v_x \sin \theta_e + v_y \cos \theta_e \\
 \dot{\theta}_e &= \omega - \frac{v_x \cos \theta_e v_y \sin \theta_e}{1 - y_e \kappa} \kappa
 \end{aligned} \tag{124}$$

The states y_e and θ_e represent the lateral error and the orientation error between the body frame and the path, respectively. These variables describe the lateral distance in (m) and heading errors in (rad) between the vehicle and the path. Additionally, κ represents the curvature of the road at each time instant, serving as the track-reference curvature.

Table 5
Dynamic model parameters for the autonomous vehicle.

Parameter	Value	Parameter	Value
l_f	0.125 m	C_f	68
l_r	0.125 m	C_r	71
m	1.98 kg	C_{dA}	0.03 m ²
I	0.03 kgm ²	g	9.8 m/s ²
ρ	1.225 kgm ³	μ	0.05

4.2.3. State of charge (SoC) of the battery

Regarding energy consumption, it is imperative to account for the State of Charge (SoC) of the vehicle's battery in the model. Essentially, SoC signifies the available energy relative to the battery's full charge capacity at a given moment. Furthermore, the operational range is typically expressed in percentage points, denoted as $\text{SoC} \in [0, 1]$, where 1 signifies a fully charged battery (100% capacity), while 0 indicates a completely discharged battery (0% capacity). According to [36], an energy storage system can be defined with regard to its discharge rate as follows:

$$\dot{\text{SoC}} = -\frac{1}{C_\alpha} p_\alpha \quad (125)$$

where p_α represents the power and C_α represents the capacity of the energy storage system, it is common practice to express p_α as a function of the vehicle's velocity, as noted in a previous reference [37].

$$p_\alpha = \frac{1}{2} C_d \rho A_r v_x^2 + \mu_s m g v_x \quad (126)$$

where C_d denotes the wheels' drag coefficient, A_r represents the vehicle's frontal cross-sectional area, and ρ stands for the air density at 25°C.

Such that the state and control vectors are as follows:

$$x = \begin{bmatrix} v_x \\ v_y \\ \omega \\ y_e \\ \theta_e \\ \text{SoC} \end{bmatrix}, u = \begin{bmatrix} a_r \\ \delta \end{bmatrix} \quad (127)$$

the discrete time non-linear model is presented as

$$x_{k+1} = f(x_k, u_k, w_k) \quad (128)$$

4.2.4. REMPC-LPV Based planner

In this article, the trajectory is planned at each discrete time step k with the dual objective of maximizing velocity and minimizing energy consumption, achieved by solving a multi-objective optimal control problem. Including the state of charge (SoC) of the battery in the vehicle model is particularly relevant in the context of electric racing cars where speed and energy efficiency are crucial for success [38]. Our innovation is to incorporate SoC into the cost function as the economic objective, ensuring the car not only finishes the laps but also maintains an adequately charged battery throughout the race.

The foremost goal is to maximize the vehicle's speed vector, allowing it to perform at its peak velocity. This is essential in racing scenarios, where achieving the highest possible speed is a key determinant of success. This objective can be formulated using the following nonlinear cost function.

$$l_v(x_{k+i}, \kappa_{k+i}) = \sum_{i=0}^{H_p} \|V_{k+i}\|_{W_v}^2 \quad (129)$$

Simultaneously, another crucial objective is to minimize energy consumption during the race. For electric racing cars, this not only ensures that the vehicle completes the laps, but also helps maintain a sufficient charge in the battery. EMPC excels in optimizing energy-efficient control actions and driving strategies, ensuring that the energy reserves are utilized judiciously over the course of the race.

In the context of our research, we introduce an economic objective function denoted as $l_e(x_{k+i}, u_{k+i})$. This objective function plays an essential role in our trajectory planning and control optimization. Its main role is to address the economic aspect of our research.

$$l_e(x_{k+i}, u_{k+i}) = \sum_{i=0}^{H_p} \|\Delta \text{SoC}_{k+i}\|_{W_e}^2. \quad (130)$$

Where W_v and W_e are weighting terms related the objectives functions l_v and l_e respectively.

4.2.5. LPV formulation

In this section, we present our innovation of transforming the nonlinear optimization problem using the LPV formulation of the nonlinear vehicle model. This novel approach offers a host of advantages over traditional methods in terms of reducing the computational load and handling the inherent challenges of parameter variation within the nonlinear model, particularly when adapting to changing operational conditions.

Additionally, to be able to represent the problem in its convex form, the nonlinear cost function of maximizing the velocity is convexified as presented in [39]. The LPV representation of the vehicle is given as follows:

$$x_{k+1} = A(\theta_k)x_k + B(\theta_k)u_k \quad (131a)$$

where the scheduling vector θ_k is chosen to be as mentioned below:

$$\theta_k := [v_x, v_y, \theta_e, \kappa, y_e, \delta]_k \quad (131b)$$

being

$$A(\theta_k) = \begin{bmatrix} A_{11} & A_{12} & A_{13} & 0 & 0 & 0 \\ 0 & A_{22} & A_{23} & 0 & 0 & 0 \\ 0 & A_{32} & A_{33} & 0 & 0 & 0 \\ 0 & A_{42} & 0 & 1 & 0 & 0 \\ -A_{51} & A_{52} & A_{53} & 0 & 0 & 0 \\ A_{61} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (131c)$$

$$B(\theta_k) = \begin{bmatrix} B_{11} & 1 \\ B_{12} & 0 \\ B_{13} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (131d)$$

where

$$\begin{aligned} A_{11} &= 1 + \left(\frac{-\mu g}{v_x} - \frac{\rho C_d \Lambda v_x}{2m} \right) T_s, A_{12} = \frac{C_f \sin \delta}{mv_x} T_s, A_{13} = \left(\frac{c_f l_f \sin \delta}{mv_x} + v_y \right) T_s, \\ A_{22} &= 1 + \left(-\frac{c_r + c_f \cos \delta}{mv_x} \right) T_s, A_{23} = \left(-\frac{c_f l_f \cos \delta - C_r l_r}{mv_x} - v_x \right) T_s, \\ A_{32} &= \frac{C_f l_f \cos \delta - l_r C_r}{I v_x} T_s, A_{33} = 1 + \left(-\frac{c_f l_f^2 \cos \delta + l_r^2 c_r}{I v_x} \right) T_s, \\ A_{42} &= \cos \theta_e T_s, A_{45} = v_x T_s, \\ A_{51} &= \frac{\kappa}{(1 - y_e \kappa)} T_s, A_{52} = \frac{\kappa}{(1 - y_e \kappa)} \sin \theta_e T_s, A_{53} = T_s, \\ A_{61} &= -\frac{1}{2C_a} p_a C_d \rho A_r v_x + \mu_s m g \\ B_{11} &= -\frac{1}{m} \sin \delta C_f \\ B_{12} &= \frac{1}{m} \cos \delta C_f \\ B_{13} &= \frac{1}{I} \cos \delta C_f l_f \end{aligned} \quad (131e)$$

At this stage, The optimization problem is not convex due to the term l_v in the objective function mentioned in (129).

In order to make it convex a convexification method is applied as mentioned in [39].

$$V^{QP} \approx V = \frac{v_x \cos \theta_e - v_y \sin \theta_e}{1 - y_e \kappa} \quad (132)$$

We consider the variable σ as a slack variable, introduced to relax the constraint on y_e as follows:

$$y_{e_{k+i}} \in [\bar{y}_e - \sigma_{k+i}, \underline{y}_e + \sigma_{k+i}] \quad (133)$$

4.2.6. Multiobjective optimization of the cost function

The two main objectives (maximizing the velocity vector and minimizing energy consumption) for the autonomous vehicle can be formulated as follows:

$$L_k = l_e(x_{k+i}, u_{k+i}) - l_v(x_{k+i}, \kappa_{k+i}) \quad (134)$$

In this formulation, the negative sign in front of l_v converts the maximization problem into a minimization one, which is more convenient for optimization algorithms.

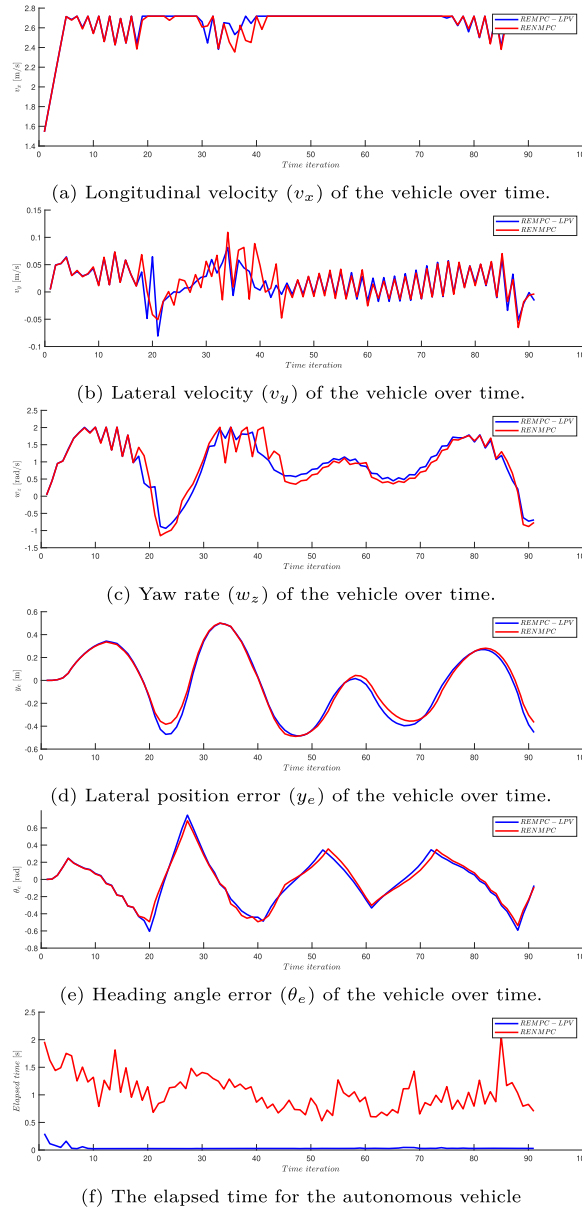


Fig. 5. Comparison of vehicle states over time using REMPC-LPV and RENMPC and optimization solver performance. These states include longitudinal velocity (v_x), lateral velocity (v_y), yaw rate (w_z), lateral position error (y_e), heading angle error (θ_e), and elapsed time for solving the optimization problem.

In this study, we implement a robust control framework for an autonomous vehicle utilizing Model Predictive Control (MPC) and Linear Parameter-Varying (LPV) control. Our approach integrates a comprehensive simulation environment that accounts for disturbances such as road slope and lateral wind, ensuring robust performance under varying conditions. By leveraging YALMIP for optimization and incorporating disturbance models, we generate realistic scenarios for autonomous driving. The control strategy employs an LPV model to handle the vehicle's nonlinear dynamics and a polytopic H-infinity controller to manage uncertainties. The simulation initializes with defined scheduling variable bounds and trajectory data, and utilizes zonotopes to manage state uncertainty. Throughout the simulation, the MPC problem is solved iteratively, updating state-space matrices and disturbance parameters to compute optimal control inputs. This methodology not only enhances the robustness of the control system but also ensures the vehicle remains within track boundaries, providing a reliable solution for autonomous vehicle navigation.

For this example the prediction horizon is set to be $H_p = 15$. The sampling time period $T_s = 0.1s$

To demonstrate the effectiveness of the proposed robust tube-based economic model predictive control method based on the linear parameter varying approach (REMPCLPV), we compare it with robust economic nonlinear MPC (RTENMPC).

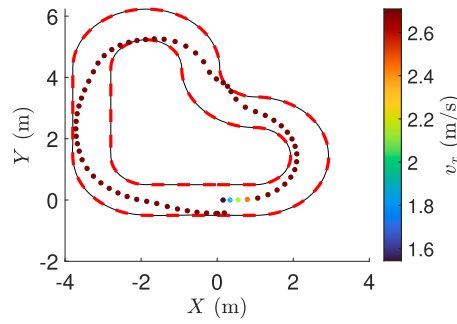


Fig. 6. Planned trajectory for one lap.

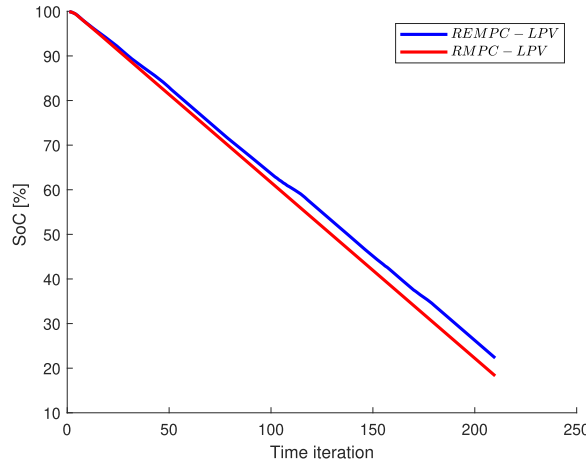


Fig. 7. SoC.

Fig. 5 depicts the vehicle states. The plot in Fig. 5a shows the longitudinal velocity v_x of the vehicle, which indicates the changes in speed along the vehicle's forward axis. The plot in Fig. 5b shows the lateral velocity (v_y), indicating side-to-side movements. The plot in Fig. 5c depicts the yaw rate (w_z), which measures the rotational speed of the vehicle around its vertical axis, relevant for turning maneuvers. The plot in Fig. 5d shows the lateral position error (e_y), quantifying the deviation from a desired path. The plot in Fig. 5e represents the heading angle error (e_θ), showing how well the vehicle maintains its desired direction. The plot in Fig. 5f shows time needed for solving the optimization problem using the two algorithms.

Fig. 6, shows the planned trajectory for one lap, as seen in the figure the vehicle reaches the maximum velocity quickly and maintain it during the whole lap, we can observe some drops in the velocity due to the curves in the map.

Fig. 7, depicts a the state of the charge of the battery using REMPC-LPV and RMPC-LPV, as observed from the plot an improvement of a 5% of the remaining SoC by using our proposed method.

The simulation results highlight a significant advantage in computational efficiency. REMPC-LPV achieves a 29-fold reduction in computational time compared to RENMPC, with mean solving times of 0.0353 s and 1.0530 s, respectively. This dramatic decrease in computational time demonstrates that REMPC-LPV is much more efficient and suitable for real-time applications where quick decision-making is crucial. The efficiency gain is likely due to the linear parameter varying approach's ability to simplify the problem structure, avoiding the complexities and iterative processes inherent in nonlinear optimization. The simulation results presented in the figures show that both REMPC-LPV and RENMPC have nearly identical responses, with REMPC-LPV exhibiting slightly better performance. This improved performance is attributed to the lack of convexity in nonlinear optimization problems.

5. Conclusions

In this paper, we introduced a robust tube-based economic model predictive control approach designed for nonlinear systems with unknown but bounded uncertainties. The proposed method employs a linear parameter varying framework to enhance the control performance and robustness. We established the input-to-state stability and recursive feasibility of the proposed approach, ensuring reliable and consistent control performance over time. By utilizing zonotopes, we efficiently handle uncertainty propagation, enabling fast and straightforward manipulation of sets. This choice enhances the robustness of the proposed method. We also incorporated a gain scheduling H-infinity controller as a local controller, which tightens the tubes along the prediction horizon, adding an extra layer of robustness to the overall control strategy. Additionally, the terminal controller used to prove the stability of the MPC controller

operates independently of the local controller, offering greater flexibility in the design process. The effectiveness of the LPV-based REMPC strategy is demonstrated through the control of a CSTR. Furthermore, an REMPC-LPV-based planning approach for a 1/10 scale autonomous RC electric car is tested through simulations, highlighting the practical applicability of the proposed method. We conducted a comparative analysis between the REMPC-LPV and its nonlinear counterpart RENMPC. The results showed significant improvements in computational time with the LPV approach. Additionally, another comparison highlighted the effectiveness of incorporating the economic objective, demonstrating improvements in the economic performance of the system. While the proposed REMPC method already demonstrates significant improvements in robustness and flexibility, it also offers an exciting opportunity for further enhancement. Currently, the estimation of scheduling variables along the prediction horizon is based on optimal values from the previous iteration, a practical and efficient approach. Building on this, future research can explore even more advanced strategies for estimating these scheduling variables to unlock additional performance gains.

CRedit authorship contribution statement

Heithem Boufrioua: Writing – original draft, Investigation; **Boubekour Boukhezzar:** Writing – review & editing; **Vicenç Puig:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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